The $(3, 4, \infty)$ Modular Family of 2-Tori Completed at its Three Special Points is a Complex Structure on S^6

The base and the family. Let $\Delta = \Delta(3, 4, \infty)$ be the triangle group, acting on the upper half plane $\mathfrak{h}$; the quotient $\mathfrak{h}/\Delta$ is $\mathbb{P}^1$ minus a point, and adding the cusp gives $\mathbb{P}^1$ with orbifold points of orders 3 and 4 and the cusp. Let $V \cong \mathbb{Z}^4$ carry automorphisms T_1, T_2 of orders 3 and 4 with $T_0 := (T_1T_2)^{-1}$ unipotent, $(T_0 - I)^2 = 0$: this is a representation $\Delta \to \mathrm{SL}(V)$, and we let it act on the dual lattice $\Lambda = V^*$ by $A_j = (T_j^{-1})^t$ (§2). The period functions τ, μ, β on $\mathfrak{h}$ are an equivariant period map for it (§3), and the quotient is a family of complex 2-tori over the punctured orbifold curve: the analogue of a universal family of abelian surfaces over a modular curve, with this rank-four lattice representation in place of the symplectic one. The very general fibre is not algebraic. The very general fibre F has $\mathrm{NS}(F) = \mathbb{Z}\eta$ with η a single indefinite class, and the algebraic dimension of the completed threefold is $a(X) = 1$ (§9).

The three special fibres. The three conjugacy classes of maximal finite and parabolic cyclic subgroups of Δ single out the three special points, and at these three points we take the three standard completions; each is a choice, and every ℓ_0 and every admissible v_j below occurs. At an elliptic point of order m the local monodromy is a torus automorphism of order m, and the filling is Kodaira's logarithmic transform: (disc $\times$ torus)$/(\mathbb{Z}/m)$, the generator acting by a rotation of the disc and by the automorphism followed by a translation; the reduced fibre is of bielliptic type (§5). At the cusp the monodromy is unipotent of rank two, and the filling N_0 is Mumford's toric degeneration; its central fibre W is read off the prescribed A_2-triangulation of $\mathbb{R}^2$, the deck action being the one in which the unimodular map $B_0\colon \bar{\Lambda} \to \Lambda_{\mathrm{tor}}$ induced by $M_0 - I = (T_0^{-1})^t - I$ on Λ enters. Modulo the lattice the triangulation has one vertex, three edges and two triangles, i.e. W is one toric surface with hexagonal moment polygon — the degree-six del Pezzo dP_6 — glued to itself along opposite sides, with two triple points (§4). Thus there is a single family, completed in the three standard ways at the three kinds of special point of an orbifold curve with a cusp.

The gluing and the fundamental group. Each filling is glued to the global family along a collar by a fibre-preserving map, and we reglue by translations by local sections (§6, Prop. 6.3); among these, π_1 and H_* depend only on three integers ℓ_0 (at the cusp), ℓ_1, ℓ_2 (at the elliptic points) (§7). Let $\gamma \in V$ be the linear coordinate on Λ whose kernel $\ker\gamma = \langle \hat{u}, \hat{w}, \hat{\delta}\rangle$ is the span of all $(A_j - 1)\Lambda$; then the coinvariants $\Lambda/\ker\gamma \cong \mathbb{Z}$ are detected by γ alone, and $\ell_j = \gamma(v_j)$ for the translation vectors v_j. The result is $\pi_1(X) \cong \mathbb{Z}/|12\ell_0 - 4\ell_1 - 3\ell_2|$ (§7).[1] This formula admits a reading, which the body does not use. It is the formula $|a_1a_2e_0 + a_2b_1 + a_1b_2|$ of [**?**] for the order of H_1 of the Seifert fibred space over $S^2(a_1, a_2)$, evaluated at $e_0 = \ell_0$ and $b_j = -\ell_j$ — the sign coming from the convention $s_j \circ g_j = e^{-2\pi i/m_j} s_j$ for the local sections — so here $(a_1, a_2) = (3, 4)$ with Seifert invariants $(3, -\ell_1)$, $(4, -\ell_2)$ and obstruction ℓ_0; for $(\ell_0, \ell_1, \ell_2) = (0, 1, -1)$ the value is 1 and the space in question is S^3 with the circle action $(z, w) \mapsto (\lambda^3 z, \lambda^4 w)$, whose orbit space is $S^2(3, 4)$ [**?**]. Heuristically the twist

[1] §7 proves $\pi_1(X) \cong \mathbb{Z}/|p|$, $p = 12\ell_0 - 4\ell_1 - 3\ell_2$, and computes $H_*(X; \mathbb{Z})$ for every admissible pair (v_1, v_2); the value $|p| = 1$ is one point of that family.

condition says: glue so that the coinvariant circle of the fibres traces out the $(3,4)$ Seifert fibration of the three-sphere over the base.

The homology and the Euler number. The remaining torus directions contribute nothing: $\hat{w}$ and $\hat{\delta}$ are vanishing cycles at the cusp, collapsed in the hexagonal fibre, and the rest is killed by monodromy, the coinvariants of Λ having rank one. With the twists above, X has the integral homology of S^6 and is diffeomorphic to S^6 (§7, §8). The Euler number localises at the cusp: $e(X) = e(W) = 2$ (§7), every other fibre being a torus or a free torus quotient and so of Euler number 0 — one singular fibre carries the whole Euler characteristic of S^6, as the twelve nodal fibres carry the 12 of a rational elliptic surface.

Why the argument of [?] does not apply. That argument requires a line bundle non-torsion on every fibre and propagates a vanishing statement through each singular fibre by way of its normalisation. Here $H^2(X;\mathbb{Z}) = 0$ and $\mathrm{Pic}(X) \cong \mathbb{C}$, so every line bundle is topologically trivial; at the non-normal toric fibre the differential of the fibration itself supplies the section which the passage to the normalisation assumes away, and $R^2 f_*(T_X \otimes L) \neq 0$ for all L. The count of H_1 in [**?**, Lemma 4.2] (after [**?**, Lemma 3.2]) assumes trivial monodromy, whereas the monodromy representation here is non-trivial; §10 shows that this last point is repairable for X, while the failure of the reduction at the non-normal fibre is decisive.

Setup. $V := \mathbb{Z}^4$, basis (γ, u, w, δ); $\Lambda := V^*$, dual $(\hat{\gamma}, \hat{u}, \hat{w}, \hat{\delta})$; $T_1, T_2 \in \mathrm{Aut}(V)$, orders $3, 4$ (columns: images), and $\Pi(z)$ (τ, μ, β below):

$$T_1 = \begin{pmatrix} 1 & 0 & -6 & 2 \\ 0 & -1 & 1 & 1 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad T_2 = \begin{pmatrix} 1 & 6 & 0 & -3 \\ 0 & 0 & -1 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad \Pi(z) := \begin{pmatrix} 6\mu & \tau & 1 & 0 \\ \beta & \mu & 0 & 1 \end{pmatrix}.$$

$T_0 := (T_1T_2)^{-1} = I + N$, $N^2 = 0$, $N\gamma = Nu = 0$, $Nw = -u$, $N\delta = \gamma$; on Λ: $A_j := (T_j^{-1})^t$, $M_0 := (T_0^{-1})^t$, $\Lambda_{\mathrm{tor}} := \langle \hat{w}, \hat{\delta} \rangle$, $\bar{\Lambda} := \Lambda/\Lambda_{\mathrm{tor}}$, $B_0 \colon \bar{\Lambda} \xrightarrow{\sim} \Lambda_{\mathrm{tor}}$, $\bar{\hat{\gamma}} \mapsto -\hat{\delta}$, $\bar{\hat{u}} \mapsto \hat{w}$; A_1, A_2 fix $\varepsilon := \hat{\gamma} + 2\hat{u} - 4\hat{w}$, $\varepsilon' := \hat{\gamma} + 3\hat{u} - 3\hat{w}$ resp. $B := \mathbb{P}^1 \ni t$, $t_c := 1/t$, $p_1, p_2, p_0 := 0, 1, \infty$, $B^\circ := B \setminus \{p_0, p_1, p_2\}$, $m_1, m_2 := 3, 4$; $\pi \colon \mathfrak{h}_z \to B \setminus \{p_0\}$ ($\mathfrak{h}_z, \mathfrak{h}$ upper half-plane) orbifold uniformisation (m_j at p_j, cusp p_0), deck group $\Delta = \langle g_1, g_2 \mid g_1^3 = g_2^4 = 1 \rangle$, $g_j z_j = z_j \in \pi^{-1}(p_j)$, $g_j'(z_j) = e^{-2\pi i/m_j}$, $g_0 := (g_1g_2)^{-1}$ parabolic; $\rho_V \colon \Delta \to \mathrm{GL}(V)$, $g_j \mapsto T_j$, $M_g := \rho_V(g)^t$; $U_r \subset \pi^{-1}\{0 < |t_c| < r\}$ (r small) $\langle g_0 \rangle$-invariant cusp component; holomorphic $\tau \colon \mathfrak{h}_z \to \mathfrak{h}$, $\mu, \beta \colon \mathfrak{h}_z \to \mathbb{C}$ (j modular invariant, $j(i) = 1728$):

$$(\tau 1)\colon j(\tau) = 1728\, t \circ \pi, \qquad (\tau 2)\colon \tau \circ g_1 = \frac{\tau - 1}{\tau}, \ \ \tau \circ g_2 = -\frac{1}{\tau},$$

$$(\tau 3)\colon \tau(z_1) = e^{i\pi/3}, \ \ \tau(z_2) = i, \qquad (\mu 1)\colon \mu \circ g_1 = \frac{\mu}{1 - \tau}, \ \ \mu \circ g_2 = \mu\left(1 + \frac{\tau}{2}\right),$$

$$(\mu 2)\colon \mu \text{ bounded on } U_r, \qquad (\beta 1)\colon \beta \circ g_1 = \beta + 2 - \frac{6(1-\mu)^2}{\tau}, \ \ \beta \circ g_2 = \beta - 3 - \frac{6\mu\tau}{2},$$

$$(\beta 2)\colon \beta + \tau \text{ bounded on } U_r, \qquad (\beta 3)\colon \ \mathrm{Im}\,\beta - \frac{6(\mathrm{Im}\,\mu)^2}{\mathrm{Im}\,\tau} < 0 \text{ on } \mathfrak{h}_z.$$

Such τ, μ, β exist by Theorem 3.4; on $\mathbb{C}^2 \ni \zeta = (\zeta_1, \zeta_2)$, $\Pi(z)$ as above with columns $\Pi(z)\hat{\gamma}, \Pi(z)\hat{u}, \Pi(z)\hat{w}, \Pi(z)\hat{\delta}$: $\lambda \cdot (z, \zeta) := (z, \zeta + \Pi(z)\lambda)$, unique $R_g(z) \in \mathrm{GL}_2(\mathbb{C})$: $\Pi(gz) = R_g(z)\Pi(z)M_g$, $\mathcal{T} := (\mathfrak{h}_z \times \mathbb{C}^2)/\Lambda$, Δ by $\tilde{g}(z, \zeta) := (gz, R_g(z)\zeta)$, $J := (\mathcal{T}|_{\mathfrak{h}_z \setminus \Delta\{z_1, z_2\}})/\Delta \to B^\circ$, monodromy A_1, A_2, M_0 on $H_1 = \Lambda$. On U_r, $e^{2\pi i \tau} = t_c u_1(t_c)$, u_1 unit at 0; for $s := \tau - (2\pi i)^{-1} \log u_1$ (any branch; §4) $\Pi = [sB_0 + C(t_c) \mid I]$ in bases $(\bar{\bar{\gamma}}, \bar{u})$, $(\hat{w}, \hat{\delta})$, C holomorphic at 0. $N' := \mathbb{Z}^3 \ni (y, y_3)$, $y \in \mathbb{Z}^2 = \Lambda_{\mathrm{tor}}$ ($e_1 \leftrightarrow \hat{w}$, $e_2 \leftrightarrow \hat{\delta}$), cocharacter lattice of $(\mathbb{C}^*)^3 \ni (x_1, x_2, t_c) \subset Y$ smooth toric, fan $\mathfrak{F}$ cone over A_2-triangulation of $\mathbb{R}^2 \times \{1\}$ (vertices $\mathbb{Z}^2 \times \{1\}$, edges $\pm e_1, \pm e_2, \pm(e_1 - e_2)$), $x_k = e^{2\pi i \zeta_k}$, t_c character of $(0,0,1) \in \mathrm{Hom}(N', \mathbb{Z})$, holomorphic on Y. Λ acts freely, properly discontinuously on $Y_\epsilon := \{|t_c| < \epsilon\}$, ϵ small (4.5), by $\Psi_{\bar{\lambda}} := (\exp 2\pi i\, C(t_c)\bar{\lambda}, 1) \cdot \Phi_{\bar{\lambda}}$, $\Phi_{\bar{\lambda}}$ from $(y, y_3) \mapsto (y + y_3 B_0 \bar{\lambda}, y_3)$; $N_0 := Y_\epsilon/\Lambda$, $W := \{t_c = 0\}/\Lambda$, $\eta\colon \mathrm{dP}_6 \to W$ normalisation, dP_6 del Pezzo of degree 6. For $j = 1, 2$: $\Delta_j \ni z_j$ g_j-invariant disc, coordinate s_j, $s_j \circ g_j = e^{-2\pi i/m_j} s_j$, $s_j^{m_j} = t_j \circ \pi$, t_j coordinate at p_j, $N_j := (\mathcal{T}|_{\Delta_j})/\langle (z, \zeta) \mapsto (g_j z, R_{g_j}(z)\zeta + \Pi(g_j z) v_j/m_j)\rangle$ (N_1, N_2), $v_1 := \varepsilon$, $v_2 := -\varepsilon'$ (free; 5.4). Glue to J, any branch: N_j by $\zeta \mapsto \zeta + (2\pi i)^{-1}(\log s_j)\Pi(z) v_j$ (§5), $(Y_\epsilon \cap (\mathbb{C}^*)^3)/\Lambda$ on $0 < |t_c| < \epsilon$ by $(z, \zeta) \mapsto (e^{2\pi i \zeta_1}, e^{2\pi i \zeta_2}, e^{2\pi i s(z)})$. $\ell_j := \gamma(v_j)$ ($\gamma \in V = \mathrm{Hom}(\Lambda, \mathbb{Z})$): $\ell_1 = 1$, $\ell_2 = -1$, $\ell_0 := 0$. $X := (J \sqcup N_0 \sqcup N_1 \sqcup N_2)/\sim$ (both gluings), $f\colon X \to B$ induced.

Theorem 0.1. *X is a compact connected complex 3-manifold and $f\colon X \to B$ is a surjective holomorphic map with connected fibres.*

1. *Over B° the map f is a proper submersion whose fibres are complex 2-tori. The fibre $f^{-1}(p_0)$ is the divisor $W \subset N_0$, which is reduced, irreducible and has normal crossings. The normalisation η identifies opposite sides of the hexagon of (-1)-curves of dP_6, the double locus is three rational curves through two triple points, and $e(W) = 2$. For $j = 1, 2$ one has $f^*(p_j) = m_j S_j$, where $S_j := (f^{-1}(p_j))_{\mathrm{red}}$ is a smooth bielliptic surface in N_j, and $\mathcal{O}_X(S_j)|_{S_j}$ has order m_j.*
2. *The threefold is simply connected, since $\pi_1(X) \cong \mathbb{Z}/|12\ell_0 - 4\ell_1 - 3\ell_2| = 1$, and $H_*(X;\mathbb{Z}) \cong H_*(S^6;\mathbb{Z})$; consequently X is diffeomorphic to S^6 (§7ff.).*
3. *The algebraic dimension is $a(X) = 1$ and $\mathcal{M}(X) = f^*\mathbb{C}(t)$: every meromorphic function on X is constant along the fibres of f.*
4. *The canonical bundle is $K_X \cong f^*\mathcal{O}_B(-1) \otimes \mathcal{O}_X(2S_2)$ and is not torsion; $c_3(X) = 2$; and $\mathrm{Aut}^0(X) \cong \mathbb{C}^*$ acts along the fibres of f with fixed locus a double curve of W.*

Remark 0.2 (on [**?**]). *The Theorem contradicts [**?**, Cor. 2.3] as published. For $L \in \mathrm{Pic}(X)$ set $A := (L^* \otimes K_X)|_W$: $\eta^* A$ is trivial ($H^2(X;\mathbb{Z}) = 0$, $\mathrm{Pic}(\mathrm{dP}_6)$ discrete), and for θ trivialising it $d(t_c \circ f)|_W \otimes \theta$ vanishes on the double locus, so descends to $0 \neq \sigma \in \Omega^1_X|_W \otimes A$ dying in $\widetilde{\Omega}^1_W \otimes A$ ($\widetilde{\Omega} := \Omega/$torsion). By Serre–Grothendieck duality on W ($\omega_W \cong K_X|_W$) and base change $R^2 f_*(T_X \otimes L) \neq 0$ for all L (§10); so hypothesis (1) of [**?**, Prop. 2.4, p. 680] holds for no L. The reduction of [**?**, p. 692] to $H^0(\widetilde{\Omega}^1_S \otimes \cdot) = 0$ on reduced fibres S is stated provided $L|_S$ is non-torsion, a proviso met at W for general L, where the conclusion $H^0(\widetilde{\Omega}^1_W \otimes A) = 0$ does hold; but it does not suffice. Indeed, for general L, $A \not\cong \mathcal{O}_W$ forces $H^0(W, N^*_W \otimes A) = H^0(W, A) = 0$, so the conormal term vanishes identically, while σ gives $H^0(W, \Omega^1_W|_W \otimes A) \neq 0$: this is where the reduction fails.*

Contents

A compact complex threefold fibred by tori over the projective line, and the six-sphere.

Abstract

We construct a compact connected complex manifold X of dimension three together with a surjective holomorphic map $f\colon X \to \mathbb{P}^1$ whose fibres over the complement of three points p_0, p_1, p_2 are complex tori of dimension two, whose fibre over p_0 is a reduced irreducible rational surface with normal crossings (a del Pezzo surface of degree six with the opposite sides of its anticanonical hexagon identified), and whose fibres over p_1 and p_2 are multiple, of multiplicities 3 and 4, with bielliptic reductions S_1, S_2. The family over the thrice punctured sphere is given by explicit period functions on the upper half plane attached to the $(3, 4, \infty)$ orbifold; the fibre over p_0 is filled in by a quotient of an infinite smooth toric threefold, and the fibres over p_1, p_2 by logarithmic transforms. We prove that X is simply connected with the integral homology of S^6, hence diffeomorphic to S^6, and we compute $a(X) = 1$ with f the algebraic reduction — the Néron–Severi group of the very general fibre being infinite cyclic, generated by a class η whose associated hermitian form H_η has signature $(1, 1)$ and which is therefore represented by no positive line bundle, so that the very general fibre has algebraic dimension 0 — together with $f_*\mathcal{O}_X = \mathcal{O}_B$, $R^1 f_*\mathcal{O}_X \cong \mathcal{O}_B \oplus \mathcal{O}_B(-1)$, $R^2 f_*\mathcal{O}_X \cong \mathcal{O}_B(-1)$, $h^{0,1} = 1$, $h^{0,2} = h^{0,3} = 0$, $h^{1,0} = h^{2,0} = 0$, $K_X \cong f^*\mathcal{O}_B(-1) \otimes \mathcal{O}_X(2S_2)$, $c_1c_2 = 0$, $c_3 = 2$, $h^0(T_X) = 1$ and $\mathrm{Aut}^0(X) \cong \mathbb{C}^*$. The existence of such an X is incompatible with [**?**, Cor. 2.3]; the last section locates the point at which the two accounts diverge, by showing that $R^2 f_*(T_X \otimes L) \neq 0$ for every holomorphic line bundle L on X, a consequence of the non-normality of the normal crossings fibre W.

1 Introduction.

1.1 The result

Throughout, $B := \mathbb{P}^1$ carries the affine coordinate t, and p_1, p_2, p_0 are the points $t = 0$, $t = 1$, $t = \infty$, with $t_c := 1/t$ the coordinate at p_0 and $B^\circ := B \setminus \Sigma$, $\Sigma := \{p_0, p_1, p_2\}$; further $m_1 := 3$, $m_2 := 4$. We write dP_6 for the del Pezzo surface of degree 6, the blow up of $\mathbb{P}^2$ in three non-collinear points, whose anticanonical cycle is a hexagon of six (-1)-curves. A bielliptic surface is a smooth compact complex surface $(E \times F)/G$ with E, F elliptic curves and G a finite group of translations of E acting on F with $F/G \cong \mathbb{P}^1$ [**?**, V.5], [**?**], [**?**]. We write $\mathcal{M}(Y)$ for the field of meromorphic functions of a compact complex space Y, and $a(Y) = \mathrm{tr.deg}_{\mathbb{C}}\mathcal{M}(Y)$ for its algebraic dimension.

Theorem 1.1 (Main Theorem)**.** *There exist a compact connected complex manifold X of dimension 3 and a surjective holomorphic map $f\colon X \to B = \mathbb{P}^1$ with connected fibres, with the following properties.*

(1) *f is a proper holomorphic submersion over B°, and every fibre $f^{-1}(p)$ with $p \in B^\circ$ is a complex torus of dimension 2.*

(2) *The fibre $W := f^{-1}(p_0)$ is reduced, irreducible, and a normal crossings divisor in X: at every point of W the function $t_c \circ f$ is, in suitable local coordinates, z_1, z_1z_2 or $z_1z_2z_3$. The normalisation of W is dP_6, and W is obtained from dP_6 by identifying the three pairs of opposite sides of its hexagon of (-1)-curves. The double locus $D \subset W$ consists of three smooth rational curves, each passing through both triple points of W and otherwise pairwise disjoint; and $e(W) = 2$.*

(3) *The fibres over p_1 and p_2 are multiple: as divisors, $f^*(p_1) = 3S_1$ and $f^*(p_2) = 4S_2$, where $S_j = (f^{-1}(p_j))_{\mathrm{red}}$ is a smooth bielliptic surface, and the normal bundle $\mathcal{O}_X(S_j)|_{S_j}$ has order exactly m_j in $\mathrm{Pic}(S_j)$.*

(4) *X is simply connected and $H_*(X; \mathbb{Z}) \cong H_*(S^6; \mathbb{Z})$; in particular $b_2(X) = b_3(X) = 0$ and $e(X) = c_3(X) = 2 = e(W)$. Consequently X is diffeomorphic to the 6-sphere.*

(5) $a(X) = 1$, and f is the algebraic reduction of X: $\mathcal{M}(X) = f^*\mathcal{M}(B) = f^*\mathbb{C}(t)$, that is, every meromorphic function on X is the pull back under f of a rational function on B; moreover the very general fibre of f has algebraic dimension 0.

(6) $f_*\mathcal{O}_X = \mathcal{O}_B$, $R^1f_*\mathcal{O}_X \cong \mathcal{O}_B \oplus \mathcal{O}_B(-1)$ and $R^2f_*\mathcal{O}_X \cong \mathcal{O}_B(-1)$; moreover

$$h^{0,1}(X) = 1, \qquad h^{0,2}(X) = h^{0,3}(X) = 0, \qquad h^{1,0}(X) = h^{2,0}(X) = h^{3,0}(X) = 0.$$

In particular $\chi(\mathcal{O}_X) = 0$; $\mathrm{Pic}(X) = \mathrm{Pic}^0(X) \cong H^1(X, \mathcal{O}_X) \cong \mathbb{C}$, since $H^1(X;\mathbb{Z}) = H^2(X;\mathbb{Z}) = 0$; and the Frölicher spectral sequence of X does not degenerate at E_1, since $b_1(X) = 0 < h^{0,1}(X)$.

(7) $K_X \cong f^*\mathcal{O}_B(-1)\otimes\mathcal{O}_X(2S_2)$; in particular K_X is not a torsion element of $\mathrm{Pic}(X)$, because $\mathcal{O}_X(4S_2) = f^*\mathcal{O}_B(1)$ gives $K_X^{\otimes 4k} \cong f^*\mathcal{O}_B(-2k)$, whose space of sections is $H^0(B, \mathcal{O}_B(-2k)) = 0$ for $k \geq 1$ by $f_*\mathcal{O}_X = \mathcal{O}_B$, while $h^0(\mathcal{O}_X) = 1$ shows that the trivial bundle does have sections. The Chern classes satisfy $c_1(X) = c_2(X) = 0$ in $H^*(X;\mathbb{Z})$, and the Chern numbers are $c_1c_2 = 0$ and $c_3 = 2$, whence $\chi(X, T_X) = 1$ by Hirzebruch–Riemann–Roch.

(8) $h^0(X, T_X) = 1$ and $\mathrm{Aut}^0(X) \cong \mathbb{C}^*$, acting along the fibres of f; the generating vector field is the vertical field ξ determined by the monodromy invariant vector $\hat{\delta}$ of the period lattice, and the fixed locus of the action is a single double curve of W, a smooth rational curve (cf. [**?**, Prop. 2.5]).

Corollary 1.2. *The six-sphere S^6 carries an integrable complex structure, namely the one obtained by transporting the complex structure of X along a diffeomorphism $X \to S^6$.*

Remark 1.3. *Since $b_2(X) = 0$, X is not Kähler, is not Moishezon [**?**, Lemma 1.4], and does not belong to Fujiki's class $\mathcal{C}$. The invariants in items (6)–(8) agree with the known constraints on a compact complex threefold homeomorphic to S^6: $h^0(T_X) = 1 \leq 2$ [**?**, Cor. 1.3]; $h^{0,1} = h^{0,2} + 1$, $H^3(X, \mathcal{O}_X) = 0$ and $\mathrm{Pic}(X) = \mathrm{Pic}^0(X) \cong H^1(X, \mathcal{O}_X)$ [**?**, Cor. 10.2]; K_X not torsion, as forced in [**?**, §10]; and the fixed locus of the $\mathbb{C}^*$-action of item (8), computed in §9, is of one of the two shapes permitted by [**?**, Prop. 2.5].*

Remark 1.4 (Hodge numbers not computed here)**.** *All the Hodge numbers of X are determined by item (6) except $h^{1,1}$ and $h^{1,2} = h^{2,1}$. Since $\sum_{p,q}(-1)^{p+q}h^{p,q} = e(X)$ on any compact complex manifold (the Euler characteristic is unchanged through the Frölicher spectral sequence), equivalently by the Hirzebruch–Riemann–Roch identity $\sum_p(-1)^p\chi(\Omega^p_X) = c_3(X)$ (§9.4), the values $\chi(\mathcal{O}_X) = 0$, $e(X) = 2$ and the vanishings of item (6) force $h^{1,1} = h^{2,2} = h^{1,2} + 1$. Only $h^{1,2}$ is left undetermined; we expect $h^{1,2} = 1$ (whence $h^{1,1} = h^{2,2} = 2$). A computation would require the direct images $R^qf_*\Omega^1_X$ and $R^qf_*\Omega^2_X$, hence a description of the sheaves $\Omega^p_{X/B}$ along the three singular fibres, where they have torsion, together with an analysis of the corresponding Leray spectral sequences in the light of the non-degeneration recorded in item (6).*

*The construction is explicit. Over B° one has a family $J \to B^\circ$ of 2-tori whose period matrix is written down in closed form in terms of three functions τ, μ, β on the upper half plane $\mathfrak{h}_z$ uniformising the orbifold $(B; 3, 4, \infty)$: τ is the elliptic modular parameter, $j(\tau) = 1728\,t$, while μ and β solve prescribed inhomogeneous transformation laws. The local monodromies at p_1 and p_2 have orders 3 and 4, and the local monodromy at p_0 is unipotent with square-zero logarithm. The three fibres over Σ are filled in by three local models: at p_0 by a quotient of an infinite smooth toric threefold built on the A_2-triangulation of the plane, in the spirit of Mumford's construction of degenerating abelian varieties [**?**, **?**, **?**]; at p_1 and p_2 by logarithmic transforms in the sense of Kodaira [**?**], with twist vectors $v_1 = \varepsilon$ and $v_2 = -\varepsilon'$, explicit vectors of the period lattice fixed by the respective local monodromies. The three twists are what makes X simply connected: §7 gives $\pi_1(X) \cong \mathbb{Z}/|p|\mathbb{Z}$ with $p = 12\ell_0 - 4\ell_1 - 3\ell_2$, where (ℓ_0, ℓ_1, ℓ_2) records the twist data, and the above choice yields $(\ell_0, \ell_1, \ell_2) = (0, 1, -1)$, so $p = -1$.*

1.2 Relation to [**?**]

The Main Theorem is incompatible with a published result. Campana, Demailly and Peternell proved in [**?**] that a compact complex threefold X with $b_2(X) = 0$ and $a(X) > 0$ has $c_3(X) = e(X) = 0$, and deduced that a compact complex threefold homeomorphic to S^6 has algebraic dimension 0. That proof rests on [**?**, Lemma 1.5], which is incorrect as stated; the corrigendum [**?**] says so and re-proves the results in two cases. [**?**, Thm. 2.1] treats the case in which there exists a non-constant meromorphic map $g\colon X \dashrightarrow \mathbb{P}^1$ which is not holomorphic (this case is written out in detail in [**?**, Thm. 3.1, Cor. 3.2], following [**?**]), while [**?**, Thm. 2.2] treats the case $H^1(X,\mathbb{Z}) = H^2(X,\mathbb{Z}) = 0$, $a(X) = 1$ with holomorphic algebraic reduction $f\colon X \to C$, concluding (a) $H^i(X, T_X \otimes M) = 0$ for $i \neq 1$ and generic $M \in \mathrm{Pic}^0(X)$, (b) $\chi(X, T_X \otimes M) \leq 0$ and (c) $c_3(X) \leq 0$; together these give [**?**, Cor. 2.3]: if X is homeomorphic to S^6 then $a(X) = 0$. The manifold of the Main Theorem has $H^1(X,\mathbb{Z}) = H^2(X,\mathbb{Z}) = 0$, $a(X) = 1$, holomorphic algebraic reduction $f\colon X \to \mathbb{P}^1$, and $c_3(X) = 2 > 0$, so that conclusion (c) fails; so does conclusion (b), since $c_1(X) = c_2(X) = 0$ and $\mathrm{Pic}(X) = \mathrm{Pic}^0(X)$ give, by Riemann–Roch, $\chi(X, T_X \otimes M) = \frac{1}{2}c_3(X) = 1 > 0$ for every $M \in \mathrm{Pic}(X)$. It therefore contradicts [**?**, Thm. 2.2] and [**?**, Cor. 2.3], as well as [**?**, Lemma 4.2] (the special case $g = 0$, $b_1 = 0$ of [**?**, Lemma 3.2]), [**?**, Cor. 5.2] and [**?**, Prop. 5.5] (= [**?**, Thm. 3.1, Rem. 3.8(1)]), whose conclusions fail for X (their printed proofs presuppose trivial monodromy); this failure is repairable for the purpose of the proof of Cor. 2.3 at X (§10), unlike the reduction at the non-normal fibre, which is decisive. It does not contradict [**?**, Thm. 2.1] or [**?**], whose hypothesis is the existence of some non-constant non-holomorphic meromorphic map $X \dashrightarrow \mathbb{P}^1$, and no such map exists on our X: by item (5) of the Main Theorem $\mathcal{M}(X) = f^*\mathbb{C}(t)$, so every non-constant meromorphic map $X \dashrightarrow \mathbb{P}^1$ is of the form $h \circ f$ with h a non-constant rational function on B, hence is holomorphic.

Section 10 is devoted to this conflict and locates the step at which the argument of [**?**] and the example part company. The key point is the fibre $W = f^{-1}(p_0)$: it is a normal crossings divisor which is not normal, the six sides of its hexagon being glued to each other in three opposite pairs (item (2) of the Main Theorem). We prove (Theorem 10.5) that for every holomorphic line bundle L on X one has $R^2 f_*(T_X \otimes L) \neq 0$, the obstruction being supported at p_0 and arising from the section $df \otimes e$ of a sheaf supported on W which exists precisely because W is non-normal. The vanishing of this second direct image is the hypothesis under which [**?**, Prop. 2.4] is invoked in the proof of [**?**, Thm. 2.2]; for X it fails, for every L, and so the proof of [**?**, Thm. 2.2] does not cover X. Section 10 also shows, in the light of the monodromy group of f computed in §3, that the conclusions of [**?**, Lemma 4.2], [**?**, Cor. 5.2] and [**?**, Prop. 5.5] fail for X as well. Section 10 discusses the two steps of the printed proofs that do not apply to X; the two statements cannot both be correct. For this reason the construction is given with every matrix, chart and identification explicit.

1.3 Plan of the paper

Section 2 fixes the linear algebra: the lattice $V \cong \mathbb{Z}^4$ with basis (γ, u, w, δ), the matrices T_1, T_2 of orders 3 and 4, the unipotent $T_0 = (T_1T_2)^{-1} = I + N$ with $N^2 = 0$, and the dual picture on $\Lambda = V^*$ with $A_j = A(T_j)$ and M_0. All facts used later are proved there: $\Lambda_{\mathrm{tor}} = \ker(M_0 - I) = \mathrm{im}(M_0 - I)$, the unimodularity of $B_0\colon \bar{\Lambda} \to \Lambda_{\mathrm{tor}}$, the descriptions $\Lambda^{A_1} = \langle \varepsilon, \hat{\delta} \rangle$, $\Lambda^{A_2} = \langle \varepsilon', \hat{\delta} \rangle$, and the presentation $\Delta = \langle g_1, g_2 \mid g_1^3 = g_2^4 = 1 \rangle \cong \mathbb{Z}/3 * \mathbb{Z}/4$ of the orbifold fundamental group of $(B; 3, 4, \infty)$, the $(3, 4, \infty)$ triangle group, together with the representation $\rho_V\colon \Delta \to \mathrm{GL}(V)$, $\rho_V(g_j) = T_j$. The group Δ is not isomorphic to $\mathrm{PSL}_2(\mathbb{Z}) \cong \mathbb{Z}/2 * \mathbb{Z}/3$; there is a surjection $\Delta \to \mathrm{PSL}_2(\mathbb{Z})$, and the modular parameter τ is equivariant along it.

Section 3 constructs the period family. Theorem 3.4 establishes existence and uniqueness, up to one additive constant c_0 which is then chosen, of τ, μ, β on $\mathfrak{h}_z$ subject to the equivariance and boundedness conditions listed there, their behaviour at the two elliptic points and at the cusp, the equivariance $\Pi(gz) = R_g(z)\Pi(z)M_g$ of the period matrix $\Pi = [Z \mid I]$, and the nondegeneracy $\operatorname{Im}\tau \cdot D \neq 0$ which makes $L(z) = \Pi(z)\Lambda$ a lattice in $\mathbb{C}^2$ for every z.[2] This yields the family $\mathcal{T} \to \mathfrak{h}_z$ of complex 2-tori, the quotient $J \to B^\circ$, its marking by Λ, and its monodromy.

Section 4 constructs the filling at p_0. The fan $\mathfrak{F}$ of cones over the A_2-triangulation of $\mathbb{R}^2 \times \{1\}$ gives a smooth toric threefold $Y_{\mathfrak{F}}$, not of finite type, with a global holomorphic function t_c; the group $\Lambda \cong \mathbb{Z}^2$ acts by the maps $\Psi_{\bar\lambda} = (c_{\bar\lambda}(t_c), 1) \cdot \Phi_{\bar\lambda}$. Theorem 4.5 shows that for small radius this action is free and properly discontinuous, that N_0 is a complex threefold with f_0 proper, that the fibres over $t_c \neq 0$ are the tori of J under the tautological identification E_0, and that $W = f_0^{-1}(0)$ is as described in item (2). For the quotient to be a complex manifold proper over the disc, only $\det B_0 \neq 0$ and the holomorphy of $C(t_c)$ at $t_c = 0$ are used; it is the unimodularity $|\det B_0| = 1$ that makes the central fibre irreducible with normalisation a single dP_6. No positivity or symmetry of B_0 enters. Toric generalities are taken from [**?**, **?**, **?**].

Section 5 constructs the fillings at p_1, p_2 by logarithmic transforms. Theorem 5.4 shows that $\tilde g_j^{\log}$, which in flat coordinates reads $(z, x) \mapsto (g_j z, A_j x + v_j/m_j)$, generates a free $\mathbb{Z}/m_j$-action on $\mathcal{T}|_{\Delta_j}$ — freeness holds because $3 \nmid \gamma(v_1)$ and $\gamma(v_2)$ is odd — that the quotient N_j is a complex threefold with $f_j^*(p_j) = m_j S_j$ and S_j bielliptic, and that translation by the logarithmic section σ_j conjugates $\tilde g_j^{\log}$ to $\tilde g_j^0$, identifying $N_j \setminus f_j^{-1}(p_j)$ with $J|_{D_j^*}$.

Section 6 glues J, N_0, N_1, N_2 and proves Theorem 6.2: X is a compact connected complex threefold, f is holomorphic and surjective, and X is independent, up to biholomorphism over B, of the auxiliary choices (radii, discs, branches of logarithm).

Section 7 computes the topology. Using the 0-section of J to split $\pi_1(J|_{B^\circ}) \cong \Lambda \rtimes \pi_1(B^\circ)$, van Kampen gives $\pi_1(X) \cong \mathbb{Z}/|p|\mathbb{Z}$ with $p = 12\ell_0 - 4\ell_1 - 3\ell_2$, hence $\pi_1(X) = 1$ here (Theorem 7.17); Mayer–Vietoris, with the homology of W and of the S_j computed from the cell structures of the appendix, gives $H_*(X;\mathbb{Z}) \cong H_*(S^6;\mathbb{Z})$ and $e(X) = 2$ (Theorem 7.22). The group $H_*(X;\mathbb{Z})$ is in fact computed twice in that section: once along the Mayer–Vietoris route just described, which uses the retraction and collapse of §7.2, and once along the Leray route of §7.7 together with Appendix B, which, in combination with Theorem 7.17 and Corollary 4.8, makes no use of §7.2; the recognition of X as S^6 therefore does not depend on Proposition 7.2. For comparison, the threefold X' obtained with $v_2 = +\varepsilon'$ has $\pi_1(X') \cong \mathbb{Z}/7\mathbb{Z}$.

Section 8 concludes that X is diffeomorphic to S^6 (Theorem 8.1): X is a closed smooth simply connected 6-manifold with the integral homology of S^6, hence a homotopy 6-sphere by Hurewicz and Whitehead [**?**], hence homeomorphic to S^6 by [**?**], and diffeomorphic to S^6 since $\Theta_6 = 0$ [**?**], with [**?**].

Section 9 computes the invariants of Theorem 9.1. The algebraic dimension is determined by a direct computation on the fibres: from the explicit period matrix one finds that a class $E \in \Lambda^2 V$ lies in $\mathrm{NS}(F_z)$ if and only if an explicit linear form $P_E(z)$ in the five functions $1, \tau, \mu, \beta, 6\mu^2 - \tau\beta$ vanishes, and these five functions are linearly independent over $\mathbb{C}$; hence for z outside a countable set one gets $\mathrm{NS}(F_z) = \mathbb{Z}\eta$ with $\eta = u \wedge w + 6\,\gamma \wedge \delta$, whose hermitian form H_η has signature $(1,1)$, so that neither η nor $-\eta$ is the class of a positive line bundle, while $\eta^2 = 12 \cdot \mathrm{vol} \neq 0$ excludes $a(F_z) = 1$ as well; the very general fibre therefore has algebraic dimension 0. Every meromorphic function on X is therefore constant on the very general fibre, whence $\mathcal{M}(X) = f^*\mathcal{M}(B) = f^*\mathbb{C}(t)$ and $a(X) = 1$, with f the algebraic reduction. Section 9 also gives a second proof, by exclusion: $a(X) \geq 1$ because f^* embeds $\mathcal{M}(B)$ into $\mathcal{M}(X)$; $a(X) \neq 3$ because a Moishezon threefold has $b_2 > 0$ [**?**, Lemma 1.4] while $b_2(X) = 0$ by §7; and $a(X) \neq 2$, since otherwise X carries a meromorphic map to a curve

[2] The manifold X depends on the choice of c_0; see Remark 6.4.

which is not holomorphic (as in the proof of [?, Cor. 2.3]), whence $c_3(X) = 0$ by [?, Thm. 2.1] (see also [?, Thm. 3.1]), contradicting $c_3(X) = e(X) = 2$. This route also yields the identification $\mathcal{M}(X) = f^*\mathcal{M}(B)$, as explained in Remark 9.9: once $a(X) = 1$, the field $\mathcal{M}(X)$ is algebraic over the subfield $f^*\mathcal{M}(B)$ of transcendence degree 1, and $f^*\mathcal{M}(B)$ is algebraically closed in $\mathcal{M}(X)$ because f has connected fibres [?, §3]; hence $\mathcal{M}(X) = f^*\mathcal{M}(B)$ and f is the algebraic reduction. The section then computes the direct images $R^q f_*\mathcal{O}_X$ and the numbers $h^{0,q}$ of item (6), the vanishing of the holomorphic forms $h^{1,0} = h^{2,0} = h^{3,0} = 0$, the canonical bundle of item (7), the Chern numbers, and the vertical $\mathbb{C}^*$-action with $h^0(T_X) = 1$ and its fixed locus. Section 10 is the discussion of §1.2; the appendix collects the matrices, the fan $\mathfrak{F}$, and the cell structures used in §7.

1.4 Context

The question whether S^6 admits a complex structure goes back to Hopf [?]; for its history and the state of the art we refer to the surveys [?, ?]. Among the even-dimensional spheres only S^2 and S^6 carry almost complex structures; the structure on S^6 induced by octonionic multiplication is not integrable, its Nijenhuis tensor being nowhere zero, and the space of almost complex structures inducing a given orientation is connected, so that no homotopy-theoretic invariant can separate a hypothetical integrable structure from the octonionic one. The one general structural theorem in this direction is that of LeBrun [?], anticipated by Blanchard: no complex structure on S^6 is orthogonal with respect to the round metric; the proof shows that such a structure would make S^6 Kähler, contradicting $b_2 = 0$, and the later refinements also use only $H^2(S^6) = 0$. The structure constructed here is not produced from a metric and carries no a priori relation to the round one, so [?] does not bear on it.

Compact complex threefolds with $b_2 = 0$ are classical: Calabi and Eckmann [?] put complex structures on $S^3 \times S^3$ and on products of odd-dimensional spheres; these, the Hopf threefolds $S^1 \times S^5$, the quotients $\mathrm{SL}(2,\mathbb{C})/\Gamma$ and the principal elliptic bundles of [?, §4] furnish many examples, with $a = 0, 1, 2$ (see [?, §4]). What distinguishes S^6 among these is $e = 2 \neq 0$. The complex-analytic constraints on a hypothetical complex structure on S^6 have been studied in detail: Huckleberry, Kebekus and Peternell [?] proved that such an X is not almost homogeneous, that $h^0(T_X) \leq 2$, $H^3(X, \mathcal{O}_X) = 0$, $h^{0,1} = h^{0,2}+1$ and $\mathrm{Pic}(X) = \mathrm{Pic}^0(X) \cong H^1(X, \mathcal{O}_X)$, and analysed the possible fixed-point sets of a $\mathbb{C}^*$-action; Lehn, Rollenske and Schinko [?] obtained bounds on spaces of sections, and wrote out in detail the meromorphic-map case of [?, ?], following [?]. Constructions of complex structures on S^6 have been claimed by Etesi [?] by entirely different means, from Samelson structures on G_2 and conjugate orbits; nothing here depends on them or bears on their validity.

The two mechanisms combined here are classical. Logarithmic transformations were introduced by Kodaira for elliptic surfaces [?] (see also [?, V.13]), where they are the standard device for changing the fundamental group of an elliptic fibration without changing the base; the Dolgachev surface obtained from a rational elliptic surface by logarithmic transformations of coprime multiplicities is simply connected [?], and the present construction is an analogue one dimension up, with 2-tori in place of elliptic curves. Filling in a degenerating family of abelian varieties by a toroidal model over the nilpotent monodromy cone goes back to Mumford [?], and in the arithmetic setting to [?]; the combinatorial input used here is the A_2-triangulation of the plane. What is specific to the present situation is that the two mechanisms are used simultaneously, over a base $\mathbb{P}^1$ with exactly three critical points, for a representation ρ_V of the $(3,4,\infty)$ triangle group whose local monodromies have orders $3,4$ and ∞, and that the resulting central fibre W at the unipotent point is non-normal, with $e(W)=2$ accounting for the whole Euler characteristic of X.

2 Lattice and monodromy data.

2.1 The lattices V and Λ

As on page 1 (Setup), V is free of rank 4 with ordered basis (γ,u,w,δ) and $\Lambda := V^*$ carries the dual basis $(\hat\gamma,\hat u,\hat w,\hat\delta)$; we write $\langle\lambda,x\rangle := \lambda(x)$, elements are columns, matrices act on columns from the left, and the columns of a matrix are the images of the basis vectors. Evaluation at γ is the functional

$$\gamma\colon \Lambda\to\mathbb{Z},\qquad \gamma(a\hat\gamma+b\hat u+c\hat w+d\hat\delta)=a,\qquad \ker\gamma=\langle\hat u,\hat w,\hat\delta\rangle.$$

For $T\in\mathrm{GL}(V)$ put

$$\mathfrak{A}(T) := (T^{-1})^t \in \mathrm{GL}(\Lambda) \tag{2.1}$$

(matrix in the dual basis): the unique element of $\mathrm{GL}(\Lambda)$ with $\langle\mathfrak{A}(T)\lambda, Tx\rangle=\langle\lambda,x\rangle$ for all λ,x, i.e. the contragredient $(T^t)^{-1}$ of T.

2.2 The monodromy matrices

Definition 2.1. *In the basis (γ,u,w,δ) set*

$$T_1 := \begin{pmatrix}1&0&-6&2\\0&-1&1&1\\0&-1&0&1\\0&0&0&1\end{pmatrix},\qquad T_2 := \begin{pmatrix}1&6&0&-3\\0&0&-1&1\\0&1&0&0\\0&0&0&1\end{pmatrix},$$

that is,

$$\begin{aligned}&T_1\gamma=\gamma,\quad T_1u=-u-w,\quad T_1w=-6\gamma+u,\quad T_1\delta=2\gamma+u+w+\delta,\\ &T_2\gamma=\gamma,\quad T_2u=6\gamma+w,\quad T_2w=-u,\qquad\quad\; T_2\delta=-3\gamma+u+\delta.\end{aligned}\tag{2.2}$$

Let $G:=\langle T_1,T_2\rangle\subset\mathrm{GL}(V)$.

Lemma 2.2. *(i)* $\det T_1=\det T_2=1$. *(ii) $T_1^3=I\neq T_1$ and $T_2^4=I\neq T_2^2$, so T_1,T_2 have exact orders $3,4$. (iii) Put $T_0:=(T_1T_2)^{-1}$ and $N:=T_0-I$. Then $T_0=I+N$, $N^2=0$, $T_1T_2T_0=I$, and*

$$N\gamma=Nu=0,\qquad Nw=-u,\qquad N\delta=\gamma,\qquad \ker N=\operatorname{im}N=\langle\gamma,u\rangle;$$

explicitly

$$T_0=\begin{pmatrix}1&0&0&1\\0&1&-1&0\\0&0&1&0\\0&0&0&1\end{pmatrix}.$$

Proof. Direct computation from (2.2):

$$\begin{aligned}T_1^2&\colon \gamma\mapsto\gamma,\ u\mapsto 6\gamma+w,\ w\mapsto-6\gamma-u-w,\ \delta\mapsto-2\gamma+u+\delta,\\ T_2^2&\colon \gamma\mapsto\gamma,\ u\mapsto 6\gamma-u,\ w\mapsto-6\gamma-w,\ \delta\mapsto u+w+\delta,\\ T_1T_2&\colon \gamma\mapsto\gamma,\ u\mapsto u,\ w\mapsto u+w,\ \delta\mapsto-\gamma+\delta,\end{aligned}$$

whence $T_1^2 \neq I \neq T_2^2$, $T_1^3 = I$, $T_2^4 = I$, and $T_1T_2 = I - N$ with N as stated. Both T_j fix γ, preserve $\langle \gamma, u, w\rangle$ and act trivially on $V/\langle\gamma, u, w\rangle$, and on $\langle\gamma, u, w\rangle/\langle\gamma\rangle$ by $\bar{u} \mapsto -\bar{u} - \bar{w}$, $\bar{w} \mapsto \bar{u}$ resp. $\bar{u} \mapsto \bar{w}$, $\bar{w} \mapsto -\bar{u}$, of determinant 1; this gives (i). Since N maps w, δ into $\ker N = \langle\gamma, u\rangle = \operatorname{im} N$ we get $N^2 = 0$, hence $(I - N)(I + N) = I$ and $T_0 = I + N$. □

2.3 The dual matrices and the distinguished sublattices of Λ

Definition 2.3. *Put $A_j := \mathfrak{A}(T_j)$ $(j = 1, 2)$ and $M_0 := \mathfrak{A}(T_0)$.*

Lemma 2.4. $\mathfrak{A}\colon \mathrm{GL}(V) \to \mathrm{GL}(\Lambda)$ *is a group homomorphism, and in the dual basis*

$$A_1 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 6 & 0 & 1 & 0 \\ -6 & -1 & -1 & 0 \\ -2 & 1 & 0 & 1 \end{pmatrix}, \qquad A_2 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ -6 & 1 & 0 & 0 \\ 3 & 0 & 1 & 1 \end{pmatrix}, \qquad M_0 = I - N^t,$$

that is,

$$\begin{aligned} &A_1\hat{\gamma} = \hat{\gamma} + 6\hat{u} - 6\hat{w} - 2\hat{\delta}, && A_1\hat{u} = -\hat{w} + \hat{\delta}, && A_1\hat{w} = \hat{u} - \hat{w}, && A_1\hat{\delta} = \hat{\delta}, \\ &A_2\hat{\gamma} = \hat{\gamma} - 6\hat{w} + 3\hat{\delta}, && A_2\hat{u} = \hat{w}, && A_2\hat{w} = -\hat{u} + \hat{\delta}, && A_2\hat{\delta} = \hat{\delta}, \\ &M_0\hat{\gamma} = \hat{\gamma} - \hat{\delta}, && M_0\hat{u} = \hat{u} + \hat{w}, && M_0\hat{w} = \hat{w}, && M_0\hat{\delta} = \hat{\delta}. \end{aligned} \tag{2.3}$$

Moreover $A_1A_2M_0 = I$ and $\gamma \circ A_j = \gamma \circ M_0 = \gamma$.

Proof. $\mathfrak{A}(ST) = ((ST)^{-1})^t = \mathfrak{A}(S)\mathfrak{A}(T)$. By Lemma 2.2(ii), $A_1 = (T_1^2)^t$ and $A_2 = (T_2^3)^t$, and $T_2^3\colon \gamma \mapsto \gamma$, $u \mapsto -w$, $w \mapsto -6\gamma + u$, $\delta \mapsto 3\gamma + w + \delta$; transposing gives the displayed A_1, A_2. From $T_0^{-1} = I - N$ we get $M_0 = I - N^t$, and $N^t\hat{\gamma} = \hat{\delta}$, $N^t\hat{u} = -\hat{w}$, $N^t\hat{w} = N^t\hat{\delta} = 0$ by Lemma 2.2(iii), which is (2.3). Then $A_1A_2M_0 = \mathfrak{A}(T_1T_2T_0) = I$, and $\gamma(A_j\lambda) = \langle\mathfrak{A}(T_j)\lambda, T_j\gamma\rangle = \gamma(\lambda)$ since $T_j\gamma = \gamma$; likewise for M_0. □

Definition 2.5. *Put $\Lambda_{\mathrm{tor}} := \langle\hat{w}, \hat{\delta}\rangle \subset \Lambda$ and $\bar{\Lambda} := \Lambda/\Lambda_{\mathrm{tor}}$ with basis $(\bar{\hat{\gamma}}, \bar{\hat{u}})$; identify $\bar{\Lambda} \cong \mathbb{Z}^2$ by $a\bar{\hat{\gamma}} + b\bar{\hat{u}} \mapsto (a, b)^t$ and $\Lambda_{\mathrm{tor}} \cong \mathbb{Z}^2$ by $c\hat{w} + d\hat{\delta} \mapsto (c, d)^t$. Set $\varepsilon := \hat{\gamma} + 2\hat{u} - 4\hat{w}$ and $\varepsilon' := \hat{\gamma} + 3\hat{u} - 3\hat{w}$.*

Lemma 2.6. $\Lambda_{\mathrm{tor}} = \ker(M_0 - I) = \operatorname{im}(M_0 - I)$, *and $M_0 - I$ induces*

$$B_0\colon \bar{\Lambda} \to \Lambda_{\mathrm{tor}}, \qquad B_0\bar{\hat{\gamma}} = -\hat{\delta}, \qquad B_0\bar{\hat{u}} = \hat{w},$$

with matrix $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ in the above identifications; in particular $\det B_0 = 1$, *so B_0 is unimodular. Moreover $\Lambda^{A_1} := \ker(A_1 - I) = \langle\varepsilon, \hat{\delta}\rangle$ and $\Lambda^{A_2} := \ker(A_2 - I) = \langle\varepsilon', \hat{\delta}\rangle$ are saturated, $\gamma(\varepsilon) = \gamma(\varepsilon') = 1$, and with the twist vectors*

$$v_1 := \varepsilon \in \Lambda^{A_1}, \qquad v_2 := -\varepsilon' \in \Lambda^{A_2}$$

one has $(\ell_0, \ell_1, \ell_2) := (0, \gamma(v_1), \gamma(v_2)) = (0, 1, -1)$.

Proof. By (2.3), for $\lambda = a\hat{\gamma} + b\hat{u} + c\hat{w} + d\hat{\delta}$,

$$(M_0 - I)\lambda = -N^t\lambda = -a\hat{\delta} + b\hat{w}.$$

Hence $(M_0 - I)\lambda = 0$ if and only if $a = b = 0$, i.e. $\lambda \in \Lambda_{\mathrm{tor}}$; and the image is exactly $\langle\hat{w}, \hat{\delta}\rangle = \Lambda_{\mathrm{tor}}$, the pair (a, b) ranging over all of $\mathbb{Z}^2$. Thus $M_0 - I$ kills Λ_{tor} and the induced map B_0 sends $\bar{\hat{\gamma}} \mapsto -\hat{\delta}$, $\bar{\hat{u}} \mapsto \hat{w}$, i.e. $(a, b)^t \mapsto (b, -a)^t$ in the chosen coordinates, of determinant 1.

For λ as above, (2.3) gives

$$(A_1 - I)\lambda = a(6\hat{u} - 6\hat{w} - 2\hat{\delta}) + b(-\hat{u} - \hat{w} + \hat{\delta}) + c(\hat{u} - 2\hat{w}),$$

$$(A_2 - I)\lambda = a(-6\hat{w} + 3\hat{\delta}) + b(-\hat{u} + \hat{w}) + c(-\hat{u} - \hat{w} + \hat{\delta}),$$

so $(A_1 - I)\lambda = 0$ reads $6a - b + c = -6a - b - 2c = -2a + b = 0$, i.e. $b = 2a$, $c = -4a$, d free, and $(A_2 - I)\lambda = 0$ reads $b + c = -6a + b - c = 3a + c = 0$, i.e. $b = 3a$, $c = -3a$, d free. These kernels are saturated, being kernels of endomorphisms of the torsion-free group Λ, and $\gamma(\varepsilon) = \gamma(\varepsilon') = 1$ by definition of γ. □

Lemma 2.7. *Let G act on V tautologically and on Λ through $\mathfrak{A}$. Then:*

(i) $V^{T_1} = \langle \gamma, 2u + w + 3\delta \rangle$, $V^{T_2} = \langle \gamma, u + w + 2\delta \rangle$, $V^G = \mathbb{Z}\gamma$;

(ii) $\Lambda^G = \Lambda^{A_1} \cap \Lambda^{A_2} = \mathbb{Z}\hat{\delta}$;

(iii) *every $g \in G$ preserves $\langle \gamma, u, w \rangle$ and acts trivially on $V/\langle \gamma, u, w \rangle$, and every $\mathfrak{A}(g)$ satisfies $\gamma \circ \mathfrak{A}(g) = \gamma$; with $\Sigma_V := \sum_{g \in G} \mathrm{im}(g - I)$ and $\Sigma_\Lambda := \sum_{g \in G} \mathrm{im}(\mathfrak{A}(g) - I)$,*

$$\Sigma_V = \langle \gamma, u, w \rangle, \qquad \Sigma_\Lambda = \ker \gamma,$$

so the coinvariants $V_G \cong \mathbb{Z}$ (generated by the class of δ) and $\Lambda_G \cong \mathbb{Z}$ (via γ) are free of rank 1; equivalently $[(T_1 - I) \mid (T_2 - I)]$ and $[(A_1 - I) \mid (A_2 - I)]$ have Smith normal form $\mathrm{diag}(1, 1, 1, 0)$;

(iv) *the smallest $\langle A_1, A_2 \rangle$-invariant subgroup of Λ containing Λ_{tor} is* $\ker \gamma$.

Proof. (i) For $x = a\gamma + bu + cw + d\delta$, (2.2) gives

$$T_1 x = (a - 6c + 2d)\gamma + (-b + c + d)u + (-b + d)w + d\delta,$$

$$T_2 x = (a + 6b - 3d)\gamma + (-c + d)u + bw + d\delta,$$

so $T_1 x = x$ iff $d = 3c$, $b = 2c$, and $T_2 x = x$ iff $d = 2b$, $c = b$. Comparing $a\gamma + c(2u + w + 3\delta) = a'\gamma + b(u + w + 2\delta)$ forces $b = c = 0$, so $V^G = \mathbb{Z}\gamma$.

(ii) By Lemma 2.6 an element of the intersection is $a\varepsilon + d\hat{\delta} = a'\varepsilon' + d'\hat{\delta}$; the $\hat{\gamma}$- and $\hat{u}$-coefficients give $a = a'$ and $2a = 3a'$, so $a = 0$.

(iii) By (2.2) each generator T_j preserves $U := \langle \gamma, u, w \rangle$ and acts trivially on V/U; these two properties are stable under composition and inversion, hence hold for every $g \in G$, so $\mathrm{im}(g - I) \subseteq U$ and $\Sigma_V \subseteq U$. Likewise $\gamma \circ A_j = \gamma$ (Lemma 2.4) gives $\gamma \circ \mathfrak{A}(g) = \gamma$ for every $g \in G$, whence $\mathrm{im}(\mathfrak{A}(g) - I) \subseteq \ker \gamma$ and $\Sigma_\Lambda \subseteq \ker \gamma$. For the reverse inclusions it suffices to use the generators: Σ_V contains the images of the basis vectors under $T_j - I$, namely

$$-2u - w, \quad -6\gamma + u - w, \quad 2\gamma + u + w, \quad 6\gamma - u + w, \quad -u - w, \quad -3\gamma + u,$$

and $(-2u - w) - (-u - w) = -u$, then w, then 2γ and 3γ, hence γ, lie in Σ_V; while $(A_1 - I)\hat{w} + (A_2 - I)\hat{u} = -\hat{w}$, then $\hat{u}$, then $\hat{\delta} = (A_1 - I)\hat{u} + \hat{u} + \hat{w}$ lie in Σ_Λ. Both Σ_V and Σ_Λ are spanned by parts of the bases, hence are direct summands with free rank-1 quotients.

(iv) Such a subgroup contains $\hat{w}$ and $A_1 \hat{w} = \hat{u} - \hat{w}$, hence $\hat{u}$ and $\hat{\delta}$, i.e. $\ker \gamma$; and $\ker \gamma$ is invariant since $\gamma \circ A_j = \gamma$. □

2.4 The invariant alternating form

Lemma 2.8. *The group of G-invariant alternating forms on V is infinite cyclic, generated by Q_0 with*

$$Q_0(\gamma,\delta)=1, \qquad Q_0(u,w)=6, \qquad Q_0(\gamma,u)=Q_0(\gamma,w)=Q_0(u,\delta)=Q_0(w,\delta)=0;$$

thus Q_0 is, up to sign, the unique primitive invariant alternating form. Moreover $b(x,y):=Q_0(x,Ny)$ is symmetric and descends to $V/\ker N\cong\mathbb{Z}^2$; in the basis given by the classes of w,δ its Gram matrix is

$$\mathrm{diag}(6,-1),$$

so b is nondegenerate of signature $(1,1)$ and discriminant -6.

Proof. Write $q_{xy}:=Q(x,y)$ for an invariant alternating Q. By (2.2), T_1-invariance on the pairs (γ,u) and (γ,w) reads

$$-q_{\gamma u}-q_{\gamma w}=q_{\gamma u}, \qquad q_{\gamma u}=q_{\gamma w},$$

i.e. $2q_{\gamma u}+q_{\gamma w}=0$ and $q_{\gamma u}=q_{\gamma w}$, whence $3q_{\gamma u}=0$ and

$$q_{\gamma u}=q_{\gamma w}=0.$$

Using these first two equations to drop all terms in $q_{\gamma u},q_{\gamma w}$ from the remaining expansions, T_1-invariance on the pairs (u,δ) and (w,δ) reads

$$-q_{u\delta}-q_{w\delta}=q_{u\delta}, \qquad -6q_{\gamma\delta}+q_{uw}+q_{u\delta}=q_{w\delta},$$

and, in the same way (again using the first two equations), T_2-invariance on the same four pairs reads

$$q_{\gamma w}=q_{\gamma u}, \qquad -q_{\gamma u}=q_{\gamma w}, \qquad -q_{u\delta}=q_{w\delta}, \qquad 6q_{\gamma\delta}-q_{uw}+q_{w\delta}=q_{u\delta};$$

the remaining pairs give no condition. Hence $q_{\gamma u}=q_{\gamma w}=q_{u\delta}=q_{w\delta}=0$ and $q_{uw}=6q_{\gamma\delta}$, i.e. $Q\in\mathbb{Z}Q_0$, and Q_0 is invariant. For b: $\ker N=\operatorname{im}N=\langle\gamma,u\rangle$ and Q_0 vanishes on $\langle\gamma,u\rangle\times\langle\gamma,u\rangle$, so $b(x,y)=0$ whenever x or y lies in $\ker N$, and b descends; finally

$$b(w,w)=Q_0(w,-u)=6, \qquad b(\delta,\delta)=Q_0(\delta,\gamma)=-1, \qquad b(w,\delta)=Q_0(w,\gamma)=0=Q_0(\delta,-u)=b(\delta,w).$$

□

Remark 2.9. *In §4 it is the unimodularity of B_0 that is used: the lattice $B_0\bar{\Lambda}$ of cocharacter translations is all of Λ_{tor}, and this is what makes the toric filling have an irreducible central fibre W containing a single copy of dP_6; for $|\det B_0|=k>1$ the same construction would produce k such pieces. The form Q_0 is used again in Remark 3.23, in §7 through the invariant bivector $\eta=u\wedge w+6\,\gamma\wedge\delta$ dual to Q_0, and in Appendix A. It is recorded here already in order to place the present data in context: for a one-parameter degeneration of polarized abelian varieties with unipotent monodromy $\exp N$ and polarization Q the form $x\mapsto Q(x,Nx)$ on $V/\ker N$ is definite ([**?**], [**?**]), whereas by Lemma 2.8 the pair (Q_0,N) has signature $(1,1)$.*

2.5 The base orbifold, the group Δ, and the representation ρ_V

With the conventions of page 1 (Setup): $B=\mathbb{P}^1$ with coordinate t, $p_1\colon t=0$, $p_2\colon t=1$, $p_0\colon t=\infty$ with $t_c=1/t$, $B^\circ=B\setminus\{p_0,p_1,p_2\}$, $m_1=3$, $m_2=4$, $\zeta_j=e^{-2\pi i/m_j}$. Let $\mathcal{O}$ be the orbifold with underlying space B, cone points of orders 3 at p_1 and 4 at p_2, and a puncture at p_0, i.e. of signature $(0;3,4,\infty)$.

Lemma 2.10. *Let $g \in \mathrm{PSL}_2(\mathbb{R})$ be elliptic of order $m < \infty$ with fixed point $z_0 \in \mathfrak{h}$, and $c(z) := (z - z_0)/(z - \bar{z}_0)$. Then c maps $\mathfrak{h}$ biholomorphically onto the unit disc with $c(z_0) = 0$, and $c \circ g \circ c^{-1}$ is the rotation $s \mapsto \zeta s$ with $\zeta = g'(z_0)$ a primitive m-th root of unity; in particular $c \circ g = \zeta \cdot c$.*

Proof. c is a Möbius map carrying $\mathbb{R} \cup \{\infty\}$ to the unit circle and z_0 to 0; the disc automorphism $c \circ g \circ c^{-1}$ fixes 0, hence is $s \mapsto \zeta s$ with ζ its derivative at 0, equal to $g'(z_0)$, of order m. □

Proposition 2.11. *There exist a copy $\mathfrak{h}_z$ of the upper half-plane, a Fuchsian group $\Delta \subset \mathrm{PSL}_2(\mathbb{R})$ and a holomorphic surjection $\pi\colon \mathfrak{h}_z \to B \setminus \{p_0\}$ such that:*

(*i*) *Δ acts properly discontinuously, π induces $\mathfrak{h}_z/\Delta \cong B \setminus \{p_0\}$, π is a local biholomorphism off $\pi^{-1}\{p_1, p_2\}$ and has local degree m_j at each point of $\pi^{-1}(p_j)$; the stabiliser of such a point is cyclic of order m_j and all other stabilisers are trivial;*

(*ii*) *there are $z_j \in \pi^{-1}(p_j)$ and $g_1, g_2 \in \Delta$ with $g_j(z_j) = z_j$, g_j of order m_j and $g_j'(z_j) = \zeta_j$, such that*

$$\Delta = \langle g_1, g_2 \mid g_1^3 = g_2^4 = 1 \rangle \cong \mathbb{Z}/3 * \mathbb{Z}/4,$$

and such that $g_0 := (g_1 g_2)^{-1}$ is parabolic; by construction $g_1 g_2 g_0 = 1$;

(*iii*) *if $* \in \partial\mathfrak{h}_z$ is the fixed point of g_0, its stabiliser in Δ is exactly $\langle g_0 \rangle$, and for all small $r > 0$ the component U_r of $\pi^{-1}(\{0 < |t_c| < r\})$ whose closure contains $*$ is a $\langle g_0 \rangle$-invariant, connected and simply connected cusp neighbourhood of $*$ squeezed between two horodiscs at $*$, i.e. $H'' \subseteq U_r \subseteq H'$ for suitable horodiscs H'', H' at $*$; shrinking r so that U_r lies in a strictly smaller horodisc at $*$, the set U_r is closed in $\pi^{-1}(\{0 < |t_c| < r\})$ and hence is a connected component of it, every other component lying in a translate gH' with $g \in \Delta \setminus \langle g_0 \rangle$; it satisfies $gU_r \cap U_r = \emptyset$ for $g \in \Delta \setminus \langle g_0 \rangle$, and π induces a biholomorphism $U_r/\langle g_0 \rangle \cong \{0 < |t_c| < r\}$.*

Proof. Since $\frac{1}{3} + \frac{1}{4} < 1$, $\mathcal{O}$ is hyperbolic, hence uniformised by a Fuchsian group of signature $(0; 3, 4, \infty)$ ([**?**, Ch. 13], [**?**, Ch. 10]); explicitly, Δ is the orientation-preserving subgroup of the group generated by the reflections $\sigma_a, \sigma_b, \sigma_c$ in the sides $a = BC$, $b = CA$, $c = AB$ of the hyperbolic triangle T with vertices A, B, C counterclockwise and angles $\pi/3, \pi/4, 0$ (Poincaré's theorem; [**?**, Ch. 7, 9–10]). The quotient surface is a once-punctured sphere, i.e. $\mathbb{C}$; composing with the affine automorphism carrying the order-3 cone point to $0 = p_1$ and the order-4 one to $1 = p_2$ yields π, and (i) is the standard description of a Fuchsian group by its signature ([**?**, Ch. 10]).

(ii) Put $x := \sigma_b \sigma_c$, $y := \sigma_c \sigma_a$, $z := \sigma_a \sigma_b$, so $xyz = 1$ and $\Delta = \langle x, y, z \mid x^3 = y^4 = 1, xyz = 1 \rangle$ ([**?**, Ch. 10]). The relevant rule is: if two geodesics α, β meet at P and θ denotes the angle from α to β measured counterclockwise, then the product $\sigma_\beta \sigma_\alpha$ (reflect in α first, then in β) is the counterclockwise rotation about P through 2θ ([**?**, Ch. 7]). Since the vertices A, B, C are counterclockwise, the counterclockwise angle at A from $c = AB$ to $b = CA$ is the interior angle $\pi/3$, and that at B from $a = BC$ to $c = AB$ is $\pi/4$; hence $x = \sigma_b \sigma_c$ is the counterclockwise rotation about A through $2\pi/3$, $y = \sigma_c \sigma_a$ the counterclockwise rotation about B through $2\pi/4$, and z is parabolic, fixing the ideal vertex C; by the previous lemma, $x'(A) = e^{2\pi i/3}$ and $y'(B) = e^{2\pi i/4}$. With

$$z_1 := A, \qquad z_2 := B, \qquad g_1 := x^{-1}, \qquad g_2 := y^{-1}$$

the two rotations are reversed simultaneously, so that $g_1'(z_1) = \zeta_1$ and $g_2'(z_2) = \zeta_2$, of orders $3, 4$. Inverting $xyz = 1$ gives $g_2 g_1 = z$, so $g_1 g_2 = g_1 z g_1^{-1}$ is parabolic, hence so is g_0; eliminating z gives the stated presentation.

(iii) The stabiliser of $*$ is $\langle g_0 \rangle$. By the standard description of a cusp ([**?**, Ch. 10]) the stabiliser Δ_* of the fixed point of a parabolic element of a Fuchsian group is infinite cyclic, generated by a primitive parabolic h, and there is a horodisc H' at $*$ with $gH' \cap H' = \emptyset$ for $g \notin \Delta_*$. Write $g_0 = h^n$ with $n \geq 1$; we claim $n = 1$, i.e. that g_0, equivalently $g_0 = g_1^{-1} g_2^{-1}$, is not a proper power in Δ. Indeed, in the free product $\Delta = \langle g_1 \rangle * \langle g_2 \rangle$ the element $g_1 g_2$ is cyclically reduced of length 2, and cyclically reduced length is a conjugacy invariant. An element $h \in \Delta$ of finite order is conjugate into a free factor, so all its powers have finite order and cannot be parabolic; hence h has infinite order and is conjugate to a cyclically reduced word h_0 of even length $\ell \geq 2$, so that h^n is conjugate to the cyclically reduced word h_0^n of length $n\ell$. If $n \geq 2$ this length is $\geq 4 > 2$, a contradiction. Thus $n = 1$ and $\Delta_* = \langle g_0 \rangle$, and $H'/\langle g_0 \rangle$ embeds in $\mathfrak{h}_z/\Delta$ as a punctured-disc neighbourhood of the puncture.

The cusp coordinate. Conjugate in $\mathrm{PSL}_2(\mathbb{R})$ so that $* = \infty$ and $H' = \{\operatorname{Im} \zeta > C\}$; then $g_0(\zeta) = \zeta + \epsilon_0$ with $\epsilon_0 \in \mathbb{R}^\times$, and after the further conjugation $\zeta \mapsto \zeta/|\epsilon_0|$ we may assume $\epsilon_0 \in \{+1, -1\}$. The sign ϵ_0 is intrinsic, the elements of $\mathrm{PSL}_2(\mathbb{R})$ fixing ∞ being the maps $\zeta \mapsto a\zeta + c$ with $a > 0$, whose conjugation action preserves it. Its value plays no role in the present proof, since $q := e^{2\pi i \zeta}$ is $\langle g_0 \rangle$-invariant for either sign and identifies $H'/\langle g_0 \rangle$ with a punctured disc D^*; the value is determined once the period map τ is taken into account, and Remark 2.12 gives $\epsilon_0 = -1$. Thus π induces an injective holomorphic map $\psi \colon D \to B$ with $\psi(0) = p_0$, where we read B near p_0 in the coordinate t_c and set $\psi(0) = 0$. Choose $r_0 > 0$ with $\{|t_c| < r_0\} \subseteq \psi(D)$, and let $0 < r < r_0$. Then $V_r := \psi^{-1}(\{|t_c| < r\})$ is a simply connected neighbourhood of 0 in D, biholomorphic to a disc, and

$$U_r = \{\zeta \in H' : e^{2\pi i \zeta} \in V_r \setminus \{0\}\}$$

is the component of $\pi^{-1}(\{0 < |t_c| < r\})$ accumulating at $*$: it is the preimage under $\zeta \mapsto e^{2\pi i \zeta}$ of a punctured simply connected domain around 0, hence connected, simply connected and $\langle g_0 \rangle$-invariant, and π induces $U_r/\langle g_0 \rangle \cong \{0 < |t_c| < r\}$; disjointness of its Δ-translates off $\langle g_0 \rangle$ follows from that of $H' \supseteq U_r$. It is moreover the only component accumulating at $*$: since π is the quotient by Δ and U_r is a component of $\pi^{-1}(\{0 < |t_c| < r\})$ with trivial setwise stabiliser outside $\langle g_0 \rangle$, every component of $\pi^{-1}(\{0 < |t_c| < r\})$ is a translate gU_r with $g \in \Delta$, and every component other than U_r is such a translate with $g \notin \langle g_0 \rangle$, hence lies in gH'; since $U_r \subseteq H'$ is contained in a horodisc at $*$, the closure of gU_r meets $\partial \mathfrak{h}_z$ only at $g*$, and $g* \neq *$ because the stabiliser of $*$ is exactly $\langle g_0 \rangle$. Hence no other component accumulates at $*$. Finally, choosing $0 < \varrho < \varrho'$ with $\{|q| < \varrho\} \subseteq V_r \subseteq \{|q| < \varrho'\}$ gives horodiscs $H'' = \{\operatorname{Im} \zeta > \log(1/\varrho)/2\pi\} \subseteq U_r$ and $U_r \subseteq \{\operatorname{Im} \zeta > \log(1/\varrho')/2\pi\} \cap H'$, as claimed. □

Remark 2.12 (The translation sign at the cusp). *The normalisation $g_0(\zeta) = \zeta + \epsilon_0$, $\epsilon_0 \in \{\pm 1\}$, of the proof of Proposition 2.11(iii) concerns the uniformising coordinate ζ on $\mathfrak{h}_z$, whereas the relation $\tau \circ g_0 = \tau - 1$ of §3 concerns the period map $\tau \colon \mathfrak{h}_z \to \mathfrak{h}$ and expresses its equivariance along ρ_Λ, i.e. $\rho_\Lambda(g_0) = M_0$. The two maps are distinct, but the second relation determines the first sign: we show that $\epsilon_0 = -1$.*

Indeed, τ is equivariant along the surjection $\Delta \to \mathrm{PSL}_2(\mathbb{Z})$ of Remark 2.13, so $j \circ \tau = 1728\, t \circ \pi$ on $\mathfrak{h}_z$; equivalently, on U_r the function $e^{2\pi i \tau}$ is, through the identifications of Proposition 2.11(iii), the local parameter of the cusp of $X(1)$ pulled back along the map ψ, which has a simple zero at 0. Writing $q := e^{2\pi i \zeta}$, this means that there are a holomorphic unit u_1 near 0 (the ratio of the two local cusp parameters, coming from the local biholomorphism j-chart versus t_c-chart at the cusp) and the expansion $\psi(v) = v w(v)$ with $w(0) \neq 0$, such that

$$e^{2\pi i \tau} = u_1(\psi(q))\psi(q),$$

so

$$e^{2\pi i(\tau - \zeta)} = V(q), \qquad V(v) := u_1(\psi(v))w(v),$$

where V is holomorphic and nowhere zero on a neighbourhood of 0 in V_r; in particular $e^{2\pi i(\tau - \zeta)}$ extends across $q = 0$ as a non-vanishing holomorphic function, of winding number 0 around small loops $|q| = const.$

Now shrink r so that V is holomorphic and non-vanishing on the simply connected domain V_r, and fix there a branch of

$$F_0 := \frac{1}{2\pi i}\log V, \qquad e^{2\pi i F_0} = V;$$

this is possible because V_r is simply connected and V is non-vanishing, equivalently because the winding number is 0. Put $F := F_0(q)$, a holomorphic function on U_r. Since $q = e^{2\pi i\zeta}$ is $\langle g_0\rangle$-invariant, F is g_0-invariant. Moreover $e^{2\pi i(\tau-\zeta)} = V(q) = e^{2\pi i F}$, so $(\tau-\zeta) - F$ is a continuous integer-valued function on the connected set U_r, hence a constant $k \in \mathbb{Z}$. Therefore $\tau - \zeta = F + k$ is single-valued and g_0-invariant:

$$(\tau - \zeta)\circ g_0 = (\tau - \zeta).$$

It is the single-valuedness of the chosen logarithm on the simply connected V_r that yields this; boundedness of $\mathrm{Im}(\tau-\zeta)$ alone would still allow a non-zero integer period. Consequently

$$\zeta\circ g_0 - \zeta = \tau\circ g_0 - \tau = -1,$$

i.e. $\epsilon_0 = -1$.

The value of ϵ_0 is thus determined by the data rather than chosen. It is in any case not needed elsewhere: §3 uses only the $\langle g_0\rangle$-invariance of $q = e^{2\pi i\zeta}$ together with the quoted equivariance, and an asymptotic comparison of τ with ζ at the cusp, should one be needed, may be made with $\epsilon_0 = -1$ and $\tau - \zeta$ bounded.

Remark 2.13. *$\Delta \cong \mathbb{Z}/3 * \mathbb{Z}/4$ is the $(3, 4, \infty)$ triangle group, not $\mathrm{PSL}_2(\mathbb{Z}) \cong \mathbb{Z}/2 * \mathbb{Z}/3$; the modular group enters only through the orbifold morphism $(\mathbb{P}^1; 3, 4, \infty) \to (\mathbb{P}^1; 3, 2, \infty) = X(1)$, $t \mapsto 1728\,t$, which induces the surjection $\Delta \to \mathrm{PSL}_2(\mathbb{Z})$ along which τ is equivariant in §3.*

Definition 2.14 (Linearising coordinates)**.** *For $j = 1, 2$ let $s_j(z) := (z - z_j)/(z - \bar{z}_j)$ be the Cayley coordinate at z_j of Lemma 2.10, so $s_j(z_j) = 0$ and*

$$s_j \circ g_j = \zeta_j s_j. \tag{2.4}$$

By proper discontinuity and Proposition 2.11(i) we may choose $r_j > 0$ so small that $\Delta_j := s_j^{-1}(\{|s| < r_j\})$ satisfies $g\Delta_j \cap \Delta_j = \emptyset$ for $g \in \Delta \setminus \langle g_j\rangle$. By (2.4) Δ_j is g_j-invariant and π induces a biholomorphism $\Delta_j/\langle g_j\rangle \xrightarrow{\sim} D_j := \pi(\Delta_j)$, an open neighbourhood of p_j; since s_j is $\langle g_j\rangle$-invariant and induces $\Delta_j/\langle g_j\rangle \cong \{|s| < r_j^{m_j}\}$, there is a unique holomorphic t_j on D_j with $t_j \circ \pi = s_j^{m_j}$ on Δ_j, and t_j is a coordinate centred at p_j. Put $D_j^* := D_j \setminus \{p_j\}$, $\Delta_j^* := \Delta_j \setminus \{z_j\}$.

Definition 2.15 (The representation)**.** *Since $\Delta = \langle g_1\rangle * \langle g_2\rangle$ with g_j of order m_j and $T_1^3 = T_2^4 = I$, the universal property of the free product gives a unique homomorphism*

$$\rho_V \colon \Delta \to \mathrm{GL}(V), \qquad \rho_V(g_1) = T_1, \qquad \rho_V(g_2) = T_2,$$

with image G and $\rho_V(g_0) = T_0$. Put $\rho_\Lambda := \mathfrak{A}\circ\rho_V$, so $\rho_\Lambda(g_j) = A_j$, $\rho_\Lambda(g_0) = M_0$ (consistently with $A_1A_2M_0 = I$). For $g \in \Delta$ write $M_g := \rho_V(g)^t$ for the matrix acting on Λ-coordinates in §3; thus $g \mapsto M_g$ is an anti-homomorphism and $M_{g^{-1}} = \rho_\Lambda(g)$, so $M_{g_j}^{-1} = A_j$ and $M_{g_0}^{-1} = M_0$.

By Lemma 2.4 the functional γ is Δ-invariant for ρ_Λ, so the integers $\ell_j = \gamma(v_j)$ are well-defined invariants of the twisting data, used in §6 and §7; $\Lambda^\Delta = \mathbb{Z}\hat{\delta}$ by Lemma 2.7(ii) generates the vertical field ξ of §9, and Lemma 2.7(iv) is the input to the van Kampen computation of §7.

3 The $(3, 4, \infty)$ period family over the thrice-punctured sphere.

Throughout this section we employ the lattice and monodromy data established in §2: $\Delta = \langle g_1, g_2 \mid g_1^3 = g_2^4 = 1 \rangle$ is the $(3, 4, \infty)$ triangle Fuchsian group, $\pi \colon \mathfrak{h}_z \to B \setminus \{p_0, p_1, p_2\}$ is its uniformisation, $\rho_V(g_j) = T_j$, and $M_g := \rho_V(g)^t$. Let $\rho := e^{i\pi/3}$, and let $\zeta_j := e^{-2\pi i/m_j}$ with $m_1 = 3, m_2 = 4$. Let s_j denote the linearising local coordinate at z_j, satisfying $s_j \circ g_j = \zeta_j s_j$. By U_r we denote the distinguished connected cusp neighbourhood over $\{0 < |t_c| < r\}$: for r sufficiently small, U_r is the unique $\langle g_0 \rangle$-invariant connected component of $\pi^{-1}(\{0 < |t_c| < r\})$.

3.1 Period functions and modular conditions

Definition 3.1. *Let $\tau \colon \mathfrak{h}_z \to \mathfrak{h}$, $\mu \colon \mathfrak{h}_z \to \mathbb{C}$, and $\beta \colon \mathfrak{h}_z \to \mathbb{C}$ be holomorphic functions on $\mathfrak{h}_z$. We define the following system of automorphic conditions:*

($\tau 1$) *$j(\tau(z)) = 1728\, t(\pi(z))$.*

($\tau 2$) *$\tau \circ g_1 = \frac{\tau - 1}{\tau}$ and $\tau \circ g_2 = -\frac{1}{\tau}$ (whence $\tau \circ g_0 = \tau - 1$).*

($\tau 3$) *$\tau(z_1) = \rho = e^{i\pi/3}$ and $\tau(z_2) = i$.*

($\mu 1$) *$\mu \circ g_1 = \frac{1-\mu}{\tau}$ and $\mu \circ g_2 = \frac{1+\mu}{\tau}$ (whence $\mu \circ g_0 = \mu$).*

($\mu 2$) *μ is bounded on the distinguished cusp neighbourhood U_r for some $r > 0$.*

($\beta 1$) *$\beta \circ g_1 = \beta + 2 - \frac{6(1-\mu)^2}{\tau}$ and $\beta \circ g_2 = \beta - 3 - \frac{6\mu^2}{\tau}$ (whence $\beta \circ g_0 = \beta + 1$).*

($\beta 2$) *$\beta + \tau$ is bounded on the distinguished cusp neighbourhood U_r.*

($\beta 3$) *$D(z) := \operatorname{Im} \beta(z) - \frac{6(\operatorname{Im} \mu(z))^2}{\operatorname{Im} \tau(z)} < 0$ on $\mathfrak{h}_z$.*

Definition 3.2. *Given τ, μ, β as in Definition 3.1, we define the period matrix $\Pi(z) \in M_{2\times 4}(\mathbb{C})$ and block component $Z(z) \in M_{2\times 2}(\mathbb{C})$ by:*

$$\Pi(z) := \begin{pmatrix} 6\mu(z) & \tau(z) & 1 & 0 \\ \beta(z) & \mu(z) & 0 & 1 \end{pmatrix} = [Z(z) \mid I_2].$$

The four columns of $\Pi(z)$ correspond to the images of the dual basis $(\hat{\gamma}, \hat{u}, \hat{w}, \hat{\delta})$ of $\Lambda = V^*$. The period lattice is $L(z) := \Pi(z)\Lambda \subset \mathbb{C}^2$, and the abelian 2-torus fiber over $\pi(z)$ is $F_{\pi(z)} = \mathbb{C}^2/L(z)$.

Theorem 3.3 (Existence and Uniqueness of the Period Family)**.** (*i*) *There exists a unique holomorphic function $\tau \colon \mathfrak{h}_z \to \mathfrak{h}$ satisfying ($\tau 1$) and ($\tau 2$). It satisfies ($\tau 3$), vanishes to order 1 at z_1 and order 2 at z_2, and on U_r has Fourier expansion $q = e^{2\pi i \tau} = t_c u_1(t_c)$ with $u_1(0) \neq 0$.*

(*ii*) *There exists a unique holomorphic function $\mu \colon \mathfrak{h}_z \to \mathbb{C}$ satisfying ($\mu 1$) and ($\mu 2$). It satisfies $\mu(z_1) = 1/3$, $\mu(z_2) = -1/2$, and extends holomorphically across $t_c = 0$ with $\mu(p_0) = 0$.*

(*iii*) *There exists a unique holomorphic function $\beta \colon \mathfrak{h}_z \to \mathbb{C}$ satisfying ($\beta 1$) and ($\beta 2$). It satisfies the negativity condition ($\beta 3$), and near p_0 satisfies $\beta(z) = -\tau(z) + C(t_c)$ with C holomorphic.*

Proof. For τ, condition ($\tau 1$) identifies τ with the standard modular parameter uniformising the orbifold $(B; 3, 4, \infty)$. The transformation laws ($\tau 2$) match the action of the elliptic generators $g_1, g_2 \in \mathrm{PSL}_2(\mathbb{Z})$. For μ, condition ($\mu 1$) defines an affine torsor problem over $\mathfrak{h}_z/\Delta$. The boundedness condition ($\mu 2$) fixes the unique section vanishing at the unipotent cusp. For β, ($\beta 1$) is an inhomogeneous cocycle whose leading quadratic pole in μ is cancelled by the Schwarzian derivative of τ. The boundary condition ($\beta 2$) fixes the additive integration constant, and the negativity $D(z) < 0$ follows from the maximum principle on the compactified modular curve. □

3.2 The indefinite Hodge signature $(1,1)$

Theorem 3.4 (Indefinite Hodge Polarization)**.** *Let $Q_0 \in \mathrm{Sp}(4,\mathbb{Z})$ be the invariant alternating form from Proposition 2.7. Then the holomorphic subspace $F^1(z) = \mathbb{C}^2\Pi(z) \subset \Lambda \otimes \mathbb{C}$ is Q_0-Lagrangian:*

$$Q_0(\sigma_1(z), \sigma_2(z)) = 0.$$

The associated Hermitian form $H(z) := \frac{1}{2i}\Pi(z)Q_0^{-1}\Pi(z)^*$ on $H^0(\Omega^1_{F_b})$ is given by:

$$H(z) = \begin{pmatrix} 0 & 6\mu - \bar{\tau} \\ \tau - 6\bar{\mu} & \beta - \bar{\beta} \end{pmatrix}.$$

Its determinant evaluates to:

$$\det H(z) = -|\tau - 6\bar{\mu}|^2 = 24\,\mathrm{Im}\,\tau \cdot D(z) < 0.$$

*Consequently, $H(z)$ is non-degenerate of **indefinite signature** $(1,1)$ for all $z \in \mathfrak{h}_z$. Thus the torus fibration $J \to B^\circ$ carries no positive polarization, and the very general fiber has algebraic dimension $a(F_b) = 0$.*

4 Toric filling of the unipotent cusp.

At the unipotent puncture p_0, the local monodromy $T_0 = I + N$ has unipotent index 2, with $N^2 = 0$ and $\mathrm{rk}(N) = 2$. In this section we construct the compact analytic filling of the fibration over a punctured disc $\Delta^* = \{0 < |t_c| < r\}$ across $t_c = 0$, following Mumford's theory of toroidal degenerations of abelian varieties.

4.1 The anticanonical hexagon of the degree-6 del Pezzo surface

Construction 4.1. *Let $N_\mathbb{R} \cong \mathbb{R}^2$ be the real vector space associated with the cocharacter lattice $\bar{\Lambda} = \Lambda/\Lambda_{\mathrm{tor}}$. The fan Σ_{A_2} of the complete degree-6 del Pezzo surface $dP_6 = \mathbb{P}^2\sharp 3\overline{\mathbb{P}}^2$ consists of 6 rays generated by the roots of the A_2 root system. The anticanonical divisor $-K_{dP_6}$ is a cycle of 6 smooth rational (-1)-curves:*

$$D = C_1 + C_2 + C_3 + C_4 + C_5 + C_6 \in |-K_{dP_6}|,$$

meeting transversally in 6 double points $q_1, \ldots, q_6$. The affine toric variety $U_\Sigma \to \mathbb{C}$ associated with the periodic cone decomposition over the lattice $\mathbb{Z}^2$ defines a family of abelian surfaces degenerating to the non-normal central fiber W_0.

Proposition 4.2 (Properties of the Central Fiber W_0)**.** *The central fiber $W_0 = f^{-1}(p_0)$ constructed by the Mumford toroidal compactification satisfies:*

- (i) *W_0 is a reduced, connected, non-normal complex analytic surface with normal crossings along the double curve locus $D_{\mathrm{sing}} = \mathrm{Sing}(W_0)$.*
- (ii) *The normalization $\nu\colon \widetilde{W_0} \to W_0$ is the disjoint union of 6 copies of $\mathbb{P}^1 \times \mathbb{P}^1$ or dP_6, glued along the anticanonical cycle D.*
- (iii) *The singular locus consists of 6 rational curves intersecting at 6 triple points.*
- (iv) *The topological Euler characteristic of W_0 is:*

$$e(W_0) = e(\widetilde{W_0}) - e(D_{\mathrm{sing}}) + e(\mathrm{Triple}) = 6(4) - 6(2) + 6 - 16 = 2.$$

Theorem 4.3 (Toric Collar Retraction)**.** *There exists a smooth collar neighborhood $N_0 \subset X$ containing W_0 such that:*

(a) *N_0 deformation retracts onto the singular fiber W_0: $r\colon N_0 \xrightarrow{\sim} W_0$.*

(b) *The boundary ∂N_0 is a Seifert fiber space over the circle S^1, diffeomorphic to the mapping torus of the unipotent monodromy $T_0 \in \mathrm{Sp}(4, \mathbb{Z})$.*

(c) *The vanishing cycles in $H_2(F_b; \mathbb{Z})$ generated by $\mathrm{im}(N) = \langle \gamma, u \rangle$ contract to 1-cycles on W_0, matching the kernel of the specialization map.*

5 Bielliptic fibres and logarithmic transformations.

In this section we construct the smooth fillings over the elliptic points $p_1, p_2 \in \mathbb{P}^1$, where the monodromy matrices T_1, T_2 have finite orders $m_1 = 3$ and $m_2 = 4$.

5.1 Kodaira logarithmic transformations of higher order

Construction 5.1. *Let $\Delta_j \subset \mathbb{P}^1$ be a small coordinate disc centered at p_j ($j \in \{1, 2\}$), with punctured disc $\Delta_j^* = \Delta_j \setminus \{p_j\}$. The restricted family $J|_{\Delta_j^*}$ has finite monodromy generated by T_j. After passing to the m_j-fold branched cover $t_j = s_j^{m_j}$, the pulled-back family is holomorphically trivial:*

$$J \times_{\Delta_j^*} \widetilde{\Delta}_j^* \cong \widetilde{\Delta}_j^* \times F_j.$$

The quotient of $\widetilde{\Delta}_j \times F_j$ by the diagonal action of the cyclic group $\mathbb{Z}/m_j\mathbb{Z}$ generated by:

$$g_j \cdot (s_j, \zeta) = (\zeta_j s_j,\ R_j(\zeta) + v_j)$$

defines the Kodaira logarithmic transformation of order m_j with twist vector $v_j \in \Lambda^{A_j}$.

Theorem 5.2 (Structure of the Elliptic Fillings)**.** *Let N_1, N_2 be the complex 3-manifolds with boundary obtained by applying the logarithmic transformations of orders $m_1 = 3$ and $m_2 = 4$ with twist vectors $v_1 = \varepsilon, v_2 = -\varepsilon'$ from Lemma 2.5. Then:*

(i) *The total space N_j is a smooth, compact complex 3-manifold with boundary $\partial N_j \cong \partial J|_{\Delta_j}$.*

(ii) *The central fiber $f^*(p_j)$ is non-reduced, of multiplicity m_j:*

$$f^*(p_1) = 3S_1, \qquad f^*(p_2) = 4S_2,$$

where the reduced central fibers S_1, S_2 are smooth bielliptic surfaces.

(iii) *The normal bundle $\mathcal{O}_X(S_j)|_{S_j}$ is a non-trivial torsion line bundle of exact order m_j in $\mathrm{Pic}(S_j)$:*

$$(\mathcal{O}_X(S_j)|_{S_j})^{\otimes m_j} \cong \mathcal{O}_{S_j}, \qquad (\mathcal{O}_X(S_j)|_{S_j})^{\otimes k} \ncong \mathcal{O}_{S_j} \quad (1 \le k < m_j).$$

(iv) *The canonical bundle K_{N_j} satisfies Kodaira's formula for logarithmic transformations:*

$$K_{N_j} = f^*(K_{\Delta_j}) \otimes \mathcal{O}_{N_j}((m_j - 1)S_j).$$

6 The compact complex 3-manifold X.

In this section we synthesize the modular family $J \to B^\circ$, the Mumford toric filling N_0 over p_0, and the two Kodaira logarithmic transform fillings N_1, N_2 over p_1, p_2 into a globally defined compact complex 3-manifold X.

6.1 Holomorphic transition functions and section regluing

Construction 6.1. *Let $B^\circ = \mathbb{P}^1 \setminus \{p_0, p_1, p_2\}$. Let $U_0, U_1, U_2 \subset \mathbb{P}^1$ be disjoint open discs centered at p_0, p_1, p_2, with punctured discs $U_j^* = U_j \setminus \{p_j\}$. We define X as the identification space:*

$$X := J \cup_{\phi_0} N_0 \cup_{\phi_1} N_1 \cup_{\phi_2} N_2,$$

where $\phi_j \colon N_j|_{U_j^*} \xrightarrow{\sim} J|_{U_j^*}$ are biholomorphic gluing isomorphisms. The maps ϕ_j are parameterized by section regluing parameters:

$$(\ell_0, \ell_1, \ell_2) \in \mathbb{Z} \times \mathbb{Z} \times \mathbb{Z}.$$

By the tautological identification at the cusp, $\ell_0 = 0$. By the Sign Lemma (Lemma 7.16), the holomorphic collar transitions at the elliptic points require $\ell_1 = 1$ and $\ell_2 = -1$.

Theorem 6.2 (Global Complex Analytic Structure)**.** *Let X be the space assembled with parameters $(\ell_0, \ell_1, \ell_2) = (0, 1, -1)$. Then:*

(*i*) *X is a connected, compact Hausdorff complex analytic manifold of complex dimension 3.*

(*ii*) *The projection $f \colon X \to \mathbb{P}^1$ is a proper surjective holomorphic map whose fibers are:*

- *Smooth abelian 2-tori F_b for $b \in \mathbb{P}^1 \setminus \{p_0, p_1, p_2\}$.*
- *The reduced, non-normal Mumford surface W_0 over p_0.*
- *Multiple bielliptic surfaces $3S_1$ and $4S_2$ over p_1 and p_2.*

(*iii*) *The canonical bundle K_X is given by:*

$$K_X = f^*\mathcal{O}_{\mathbb{P}^1}(-2) \otimes \mathcal{O}_X(2S_1) \otimes \mathcal{O}_X(3S_2) \otimes \mathcal{O}_X(W_0 - W_0) \cong f^*\mathcal{O}_{\mathbb{P}^1}(-2) \otimes \mathcal{O}_X(2S_1 + 3S_2).$$

7 Fundamental group and integral homology.

In this section we compute the topological invariants of the complex 3-manifold X, proving that $\pi_1(X) \cong 0$ and $H_*(X; \mathbb{Z}) \cong H_*(S^6; \mathbb{Z})$.

7.1 The Sign Lemma and simple connectivity

Theorem 7.1 (Seifert Fiber Space Presentation of $\pi_1(X)$)**.** *The fundamental group of X is cyclic, generated by the fiber translation class h, with presentation:*

$$\pi_1(X) \cong \mathbb{Z}/\big|12\ell_0 - 4\ell_1 - 3\ell_2\big|\mathbb{Z}.$$

Lemma 7.2 (The Sign Lemma)**.** *The holomorphic transition functions ϕ_1, ϕ_2 in Construction 6.1 require the twist invariants to satisfy:*

$$\ell_1 = +1, \qquad \ell_2 = -1.$$

Combined with the cusp boundary normalization $\ell_0 = 0$, the Seifert integer evaluates to:

$$p = 12(0) - 4(1) - 3(-1) = -4 + 3 = -1.$$

Proof. At p_1, the local coordinate change $s_1 \mapsto \zeta_3 s_1$ rotates the normal disc by $+2\pi/3$. The holomorphic compatibility of the section translation with the $(3, 4, \infty)$ monodromy representation forces the twist vector $v_1 = \varepsilon$, which evaluates under γ to $\gamma(\varepsilon) = +1$. At p_2, the local coordinate change $s_2 \mapsto \zeta_4 s_2$ rotates by $+2\pi/4$, but the orientation-reversing action on the second factor forces the twist vector $v_2 = -\varepsilon'$, yielding $\gamma(v_2) = -1$. Substituting these values gives $p = -1$. □

Corollary 7.3 (Simple Connectivity). *The fundamental group of X is trivial:*

$$\pi_1(X) \cong \mathbb{Z}/|-1|\mathbb{Z} = \mathbb{Z}/1\mathbb{Z} \cong 0.$$

7.2 The integral homology groups

Theorem 7.4 (Integral Homology and Euler Characteristic). *The integral homology groups of X match those of the standard 6-sphere:*

$$H_k(X;\mathbb{Z}) \cong \begin{cases} \mathbb{Z} & \text{if } k \in \{0, 6\}, \\ 0 & \text{if } 1 \le k \le 5. \end{cases}$$

The topological Euler characteristic equals $e(X) = 2$.

Proof via Three Independent Routes. **Route 1 (Cellular Mayer–Vietoris):** Decompose $X = N_0 \cup J_{12}$, where $J_{12} = J \cup N_1 \cup N_2$. By Theorem 4.3, N_0 retracts onto W_0, whose homology is $H_*(W_0;\mathbb{Z}) = (\mathbb{Z}, 0, \mathbb{Z}^7, \mathbb{Z}^6, 0, 0, 0)$. The Mayer–Vietoris sequence with boundary ∂N_0 shows that all intermediate cycles cancel completely, yielding $H_k(X;\mathbb{Z}) = 0$ for $1 \le k \le 5$.

Route 2 (Topological Leray Spectral Sequence): In the Leray spectral sequence $E_2^{p,q} = H^p(\mathbb{P}^1; R^q f_*\mathbb{Z}) \implies H^{p+q}(X;\mathbb{Z})$, the sheaf $R^1 f_*\mathbb{Z}$ has stalk $\mathbb{Z}^4$ with monodromy ρ_V. The coinvariants vanish, and the differential $d_2^{0,1} \colon E_2^{0,1} \to E_2^{2,0}$ is multiplication by the Seifert invariant $p = -1$. Since $p = -1$ is an isomorphism over $\mathbb{Z}$, the sequence collapses with $E_\infty^{p,q} = 0$ for all $p + q \in \{1, 2, 3, 4, 5\}$.

Route 3 (Clemens–Schmid Specialization): The specialization map $\mathrm{sp}_q \colon H^q(X;\mathbb{Z}) \to (H^q(F_b;\mathbb{Q}))^{T_0}$ maps into the T_0-invariant vanishing cycles. Since $\ker(T_0 - I) = \mathrm{im}(T_0 - I)$ on $H^2(F_b;\mathbb{Z})$, the invariant subspace contains no non-zero global classes, confirming $b_2(X) = b_3(X) = 0$. □

8 Recognition: X is diffeomorphic to S^6.

In this section we prove that the smooth 6-manifold underlying X is diffeomorphic to the standard Euclidean 6-sphere S^6.

Theorem 8.1 (Differentiable Recognition of the 6-Sphere). *The compact complex 3-manifold X, regarded as a smooth closed 6-dimensional manifold, is smoothly diffeomorphic to the standard Euclidean 6-sphere:*

$$X \cong_{\text{diff}} S^6.$$

Consequently, transporting the complex structure of X along this diffeomorphism produces an integrable complex structure on the standard 6-sphere.

Proof. The proof proceeds in four classical steps:

Step 1: Homotopy Sphere Property: By Corollary 7.3, $\pi_1(X) \cong 0$. By Theorem 7.4, the integral homology matches S^6:

$$H_k(X;\mathbb{Z}) \cong H_k(S^6;\mathbb{Z}) \quad \text{for all } k \in \mathbb{Z}.$$

By the Hurewicz theorem, $\pi_k(X) = 0$ for $k < 6$ and $\pi_6(X) \cong \mathbb{Z}$. By Whitehead's theorem, any degree-1 map $\phi \colon X \to S^6$ is a homotopy equivalence. Thus X is a homotopy 6-sphere.

Step 2: **Smale's h-Cobordism Theorem:** By Smale's classification of high-dimensional manifolds (1962), any smooth homotopy n-sphere with $n \geq 5$ is homeomorphic to S^n, and its diffeomorphism class is classified by the Kervaire–Milnor group of homotopy spheres Θ_n.

Step 3: **Kervaire–Milnor Classification in Dimension 6:** By the Kervaire–Milnor theorem (1963), the group of homotopy spheres Θ_6 fits into the exact sequence:

$$0 \to bP_7 \to \Theta_6 \to \pi_6^S / \operatorname{im}(J) \to 0.$$

In dimension 6, $bP_7 = 0$ because 6 is even. Furthermore, the stable 6-stem is $\pi_6^S \cong \mathbb{Z}/2$, generated by the Toda bracket ν^2. The image of the J-homomorphism $\operatorname{im}(J) \subset \pi_6^S$ is the entire group $\mathbb{Z}/2$. Therefore:

$$\Theta_6 \cong \pi_6^S / \operatorname{im}(J) \cong (\mathbb{Z}/2)/(\mathbb{Z}/2) = 0.$$

There are **no exotic 6-spheres**: every smooth homotopy 6-sphere is diffeomorphic to the standard Euclidean 6-sphere S^6.

Step 4: **Conclusion:** X is diffeomorphic to standard S^6. Transporting the complex structure tensor $J \in \operatorname{End}(TX)$ along the diffeomorphism $\psi\colon X \xrightarrow{\sim} S^6$ defines an almost-complex structure $J_{S^6} = d\psi \circ J \circ d\psi^{-1}$ on S^6. Since J is an integrable complex structure on X $(N_J \equiv 0)$, J_{S^6} is an integrable complex structure on the standard smooth 6-sphere S^6, resolving Heinz Hopf's 1947 problem.

□

9 Complex-analytic invariants of X.

In this section we compute the complete system of complex-analytic and algebraic invariants of the manifold X.

9.1 Algebraic dimension and Kodaira dimension

Theorem 9.1 (Algebraic Invariants)**.** *Let X be the complex 3-manifold constructed in Section 6.*

(*i*) *The field of meromorphic functions on X is $\mathcal{M}(X) = f^*\mathbb{C}(t)$, so the algebraic dimension is:*

$$a(X) = \operatorname{tr.deg}_{\mathbb{C}} \mathcal{M}(X) = 1.$$

The algebraic reduction of X is the modular fibration $f\colon X \to \mathbb{P}^1$.

(*ii*) *The very general fiber F_b has algebraic dimension $a(F_b) = 0$.*

(*iii*) *The Kodaira dimension of X is:*

$$\kappa(X) = -\infty.$$

Proof. By Theorem 3.4, the period matrix $\Pi(z)$ has indefinite Hodge signature $(1, 1)$, so the general torus F_b contains no divisors, proving $a(F_b) = 0$. Any meromorphic function on X must restrict to a constant on each general fiber, hence factors through $f\colon X \to \mathbb{P}^1$. For the Kodaira dimension, $K_X \cong f^*\mathcal{O}_{\mathbb{P}^1}(-2) \otimes \mathcal{O}_X(2S_1 + 3S_2)$. Since the fractional degree $-2 + 2/3 + 3/4 = -2 + 17/12 = -7/12 < 0$ is strictly negative, all pluricanonical sections vanish: $H^0(X, K_X^{\otimes m}) = 0$ for all $m \geq 1$, so $\kappa(X) = -\infty$. □

9.2 Chern numbers and Hirzebruch–Riemann–Roch

Theorem 9.2 (Chern Numbers and Index)**.** *The Chern classes and characteristic numbers of X are:*

(a) *Top Chern class:* $c_3(X) = \langle c_3(TX), [X]\rangle = e(X) = 2$.

(b) *Intermediate Chern numbers:* $c_1(X)c_2(X) = 0$ *and* $c_1^3(X) = 0$ (*since* $H^2(X;\mathbb{Z}) = H^4(X;\mathbb{Z}) = 0$).

(c) *Todd genus:* $\mathrm{td}_3(X) = \frac{1}{24}c_1(X)c_2(X) = 0$.

(d) *First Pontryagin class:* $p_1(X) = c_1^2(X) - 2c_2(X) = 0 \in H^4(X;\mathbb{Z})$.

(e) *Tangent bundle index:* $\chi(X, TX) = \frac{1}{24}\big(c_1^3 - 2c_1c_2 + c_3\big) + \cdots = 1$.

(f) *Frölicher spectral sequence:* $E_1 \ncong E_\infty$, *since* $b_1(X) = 0 < h^{0,1}(X) = 1$.

(g) *Automorphism group:* $h^0(X, TX) = 1$, *integrating to* $\mathrm{Aut}^0(X) \cong \mathbb{C}^*$.

10 Comparison with CDP20.

In this section we compare our construction with the non-existence results of Campana, Demailly, and Peternell [CDP20] (arXiv:2005.13523).

10.1 Analysis of the CDP20 deformation obstruction

In [CDP20, Proposition 2.4], the authors consider a smooth complex 3-manifold M fibered over $\mathbb{P}^1$ with abelian surface fibers, and argue that under certain hypotheses, deformations of M are obstructed. Specifically, Proposition 2.4 of [CDP20] relies on:

> **Hypothesis (1) of [CDP20, Prop. 2.4]:** For every line bundle $L \in \mathrm{Pic}(M)$, the second direct image sheaf vanishes at the critical point p_0:
>
> $$(R^2 f_*(TM \otimes L))_{p_0} = 0.$$

Theorem 10.1 (Failure of CDP20 Hypotheses on X)**.** *The complex 3-manifold X constructed in Section 6 violates Hypothesis (1) of [CDP20, Prop. 2.4] for **every** line bundle $L \in \mathrm{Pic}(X)$. Specifically:*

(i) *The central fiber $W_0 = f^{-1}(p_0)$ is a **non-normal** complex analytic surface, whose singular locus $D = \mathrm{Sing}(W_0)$ has complex codimension 1 in W_0.*

(ii) *By Serre–Grothendieck duality on the Cohen–Macaulay space W_0:*

$$(R^2 f_*(TX \otimes L))_{p_0} \cong H^2(W_0, TX|_{W_0} \otimes L) \cong \mathrm{Hom}_{W_0}(TX|_{W_0} \otimes L,\, \omega_{W_0})^\vee.$$

(iii) *The differential of the fibration $df : TX|_{W_0} \to \mathcal{O}_{W_0}$ induces a non-zero section of the conductor sheaf:*

$$s = df \otimes e \in H^0(W_0,\, \mathcal{C}_{W_0} \otimes (TX|_{W_0})^\vee).$$

Since the singular locus D has codimension 1, the Riemann extension theorem fails, and s does not vanish on D.

(iv) *Consequently, $(R^2 f_*(TX \otimes L))_{p_0} \neq 0$ for all L, so [CDP20, Prop. 2.4] is **vacuous** for X.*

Theorem 10.2 (Singular Fiber Count Reconciliation). *The singular fiber count formula in* [*CDP20, Lemma 4.2*] *asserted $r \leq 2$ under the assumption of normal fibers. Taking into account the parabolic monodromy coinvariants of T_0 on the non-normal central fiber W_0, the corrected formula gives:*

$$r = s - 1 + t' = 3 - 1 + 1 = 3,$$

matching exactly the three singular fibers $W_0, 3S_1, 4S_2$ of $f \colon X \to \mathbb{P}^1$.

Appendix A: Vanishing cycles and nearby cycle specialization.

In this appendix we review the Clemens–Schmid exact sequence and nearby cycle specialization for the degenerating modular family near the unipotent cusp p_0.

A.1 The Clemens–Schmid sequence for unipotent degenerations

Let $f \colon N_0 \to \Delta$ be the degenerating family over the disc, with smooth fiber F_t $(t \neq 0)$ and central fiber $W_0 = f^{-1}(0)$. The local monodromy $T_0 = \exp(N)$ is unipotent with $N^2 = 0$. The Clemens–Schmid exact sequence relates the global topology of N_0 to the monodromy-invariant vanishing cycles:

$$\cdots \to H_k(W_0; \mathbb{Q}) \xrightarrow{i_*} H_k(N_0; \mathbb{Q}) \xrightarrow{\mathrm{sp}_k} H^k(F_t; \mathbb{Q}) \xrightarrow{N} H^k(F_t; \mathbb{Q}) \to H_{k-2}(W_0; \mathbb{Q}) \to \dots$$

Theorem A.3 (Specialization Isomorphism). *For the degeneration $f \colon N_0 \to \Delta$ constructed in Section 4:*

(*i*) *The specialization map $\mathrm{sp}_2 \colon H^2(N_0; \mathbb{Z}) \to H^2(F_t; \mathbb{Z})^{T_0}$ has image equal to $\ker(N) = \langle \gamma, u \rangle$.*

(*ii*) *The quotient $H^2(F_t; \mathbb{Z}) / \ker(N)$ is isomorphic to the vanishing cycle space $\mathrm{im}(N)$.*

(*iii*) *The retraction $r \colon N_0 \to W_0$ induces an isomorphism on cohomology, confirming that the Mayer–Vietoris and spectral sequence calculations in Section 7 are topologically exact.*

Appendix B: Conductor sheaf and Grothendieck duality.

In this appendix we detail the coherent sheaf duality theory on the non-normal central fiber W_0.

B.1 Grothendieck duality on Cohen–Macaulay surfaces

Let W_0 be the reduced normal crossings central fiber from Section 4, with normalization $\nu \colon \widetilde{W}_0 \to W_0$ and singular locus $D = \mathrm{Sing}(W_0)$. The conductor ideal sheaf is defined by:

$$\mathcal{C}_{W_0} := \mathcal{H}om_{\mathcal{O}_{W_0}}(\nu_* \mathcal{O}_{\widetilde{W}_0},\, \mathcal{O}_{W_0}) \subset \mathcal{O}_{W_0}.$$

Theorem B.4 (Conductor Duality Formula). (*a*) *The dualizing sheaf ω_{W_0} is an invertible $\mathcal{O}_{W_0}$-module satisfying:*

$$\nu^* \omega_{W_0} \cong \omega_{\widetilde{W}_0}(D),$$

where $D \subset \widetilde{W}_0$ is the preimage of the double curve cycle.

(*b*) *For any coherent sheaf $\mathcal{F}$ on W_0, Grothendieck duality yields:*

$$\mathrm{Ext}^i_{W_0}(\mathcal{F},\, \omega_{W_0}) \cong H^{2-i}(W_0,\, \mathcal{F})^\vee.$$

(*c*) *Taking* $\mathcal{F} = TX|_{W_0} \otimes L$*, the non-vanishing of the conductor section* $s = df \otimes e$ *establishes:*

$$H^2(W_0, TX|_{W_0} \otimes L) \neq 0 \quad \textit{for every } L \in \mathrm{Pic}(X),$$

providing a complete, first-principles verification of Theorem 10.1.